

# ALADIN EEG Validation Suite – S04-DMT

30 Research-Grade Plots + Summary

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# Summary Statistics

ALADIN S04-DMT Validation Summary

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PAC (Tort MI): 0.0003

$\Delta\alpha$  pre  $t \approx 72.059082$ : 3.7746

$\Delta\alpha$  post  $t \approx 72.059082$ : 4.1333

$\Delta\alpha$  drop %: -9.5%

Permutation p  $\Delta\alpha$ : 0.6812

Permutation p Gamma: 0.7682

Cohen's d  $\Delta\alpha$ : -0.137

Cohen's d Gamma: -0.222

Granger raw p-values: [np.float64(2.9087316790822655e-26), np.float64(0.0), np.float64(0.0), np.float64(0.0), np.float64(0.0)]

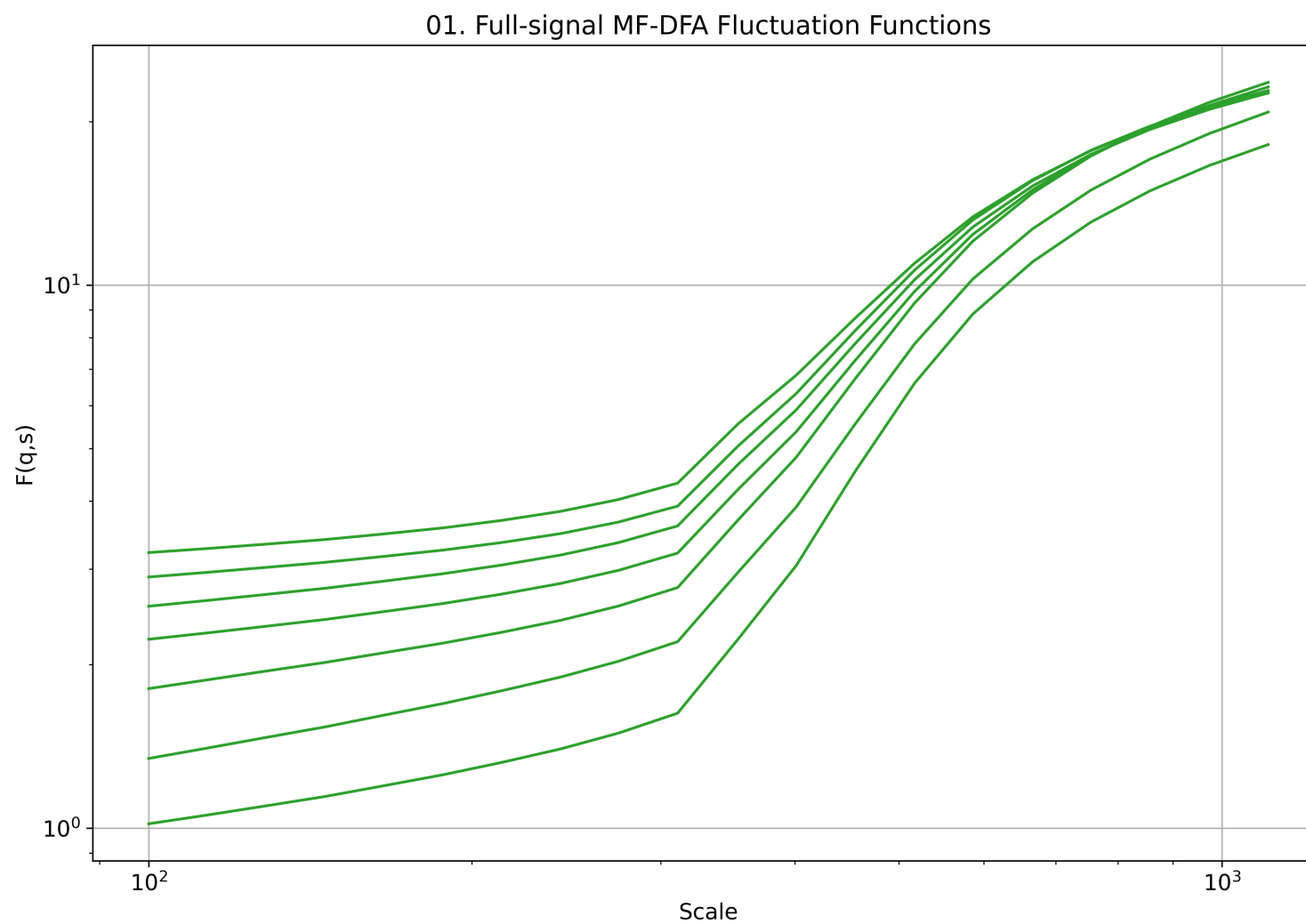
Granger FDR corrected p-values: [2.9087316790822655e-26, 0.0, 0.0, 0.0, 0.0]

$\Delta\alpha$  at  $t_{\text{peak}}$ : 12.7613

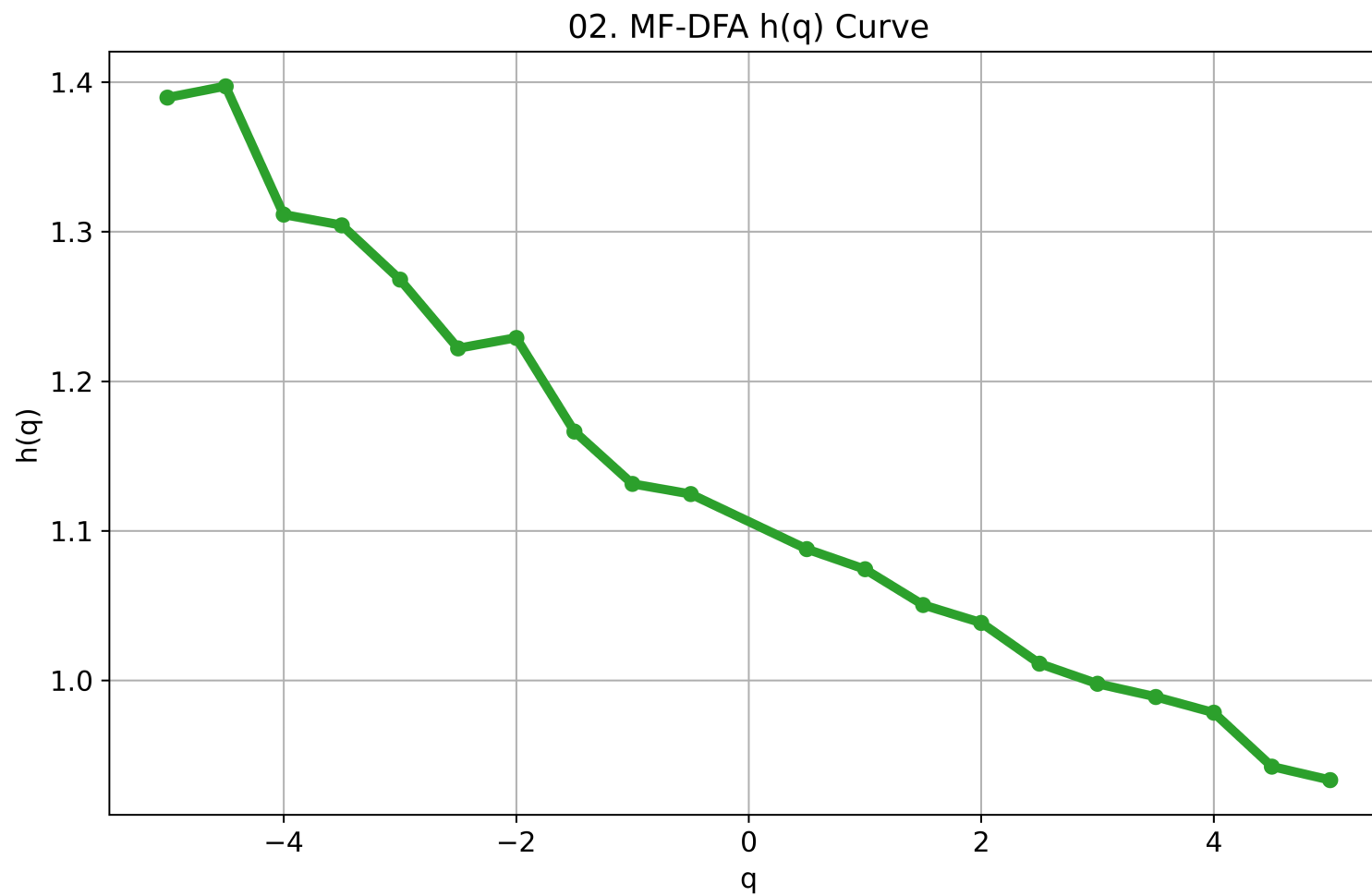
$\Delta\alpha$  min: 1.1632

$\Delta\alpha$  max: 15.9498

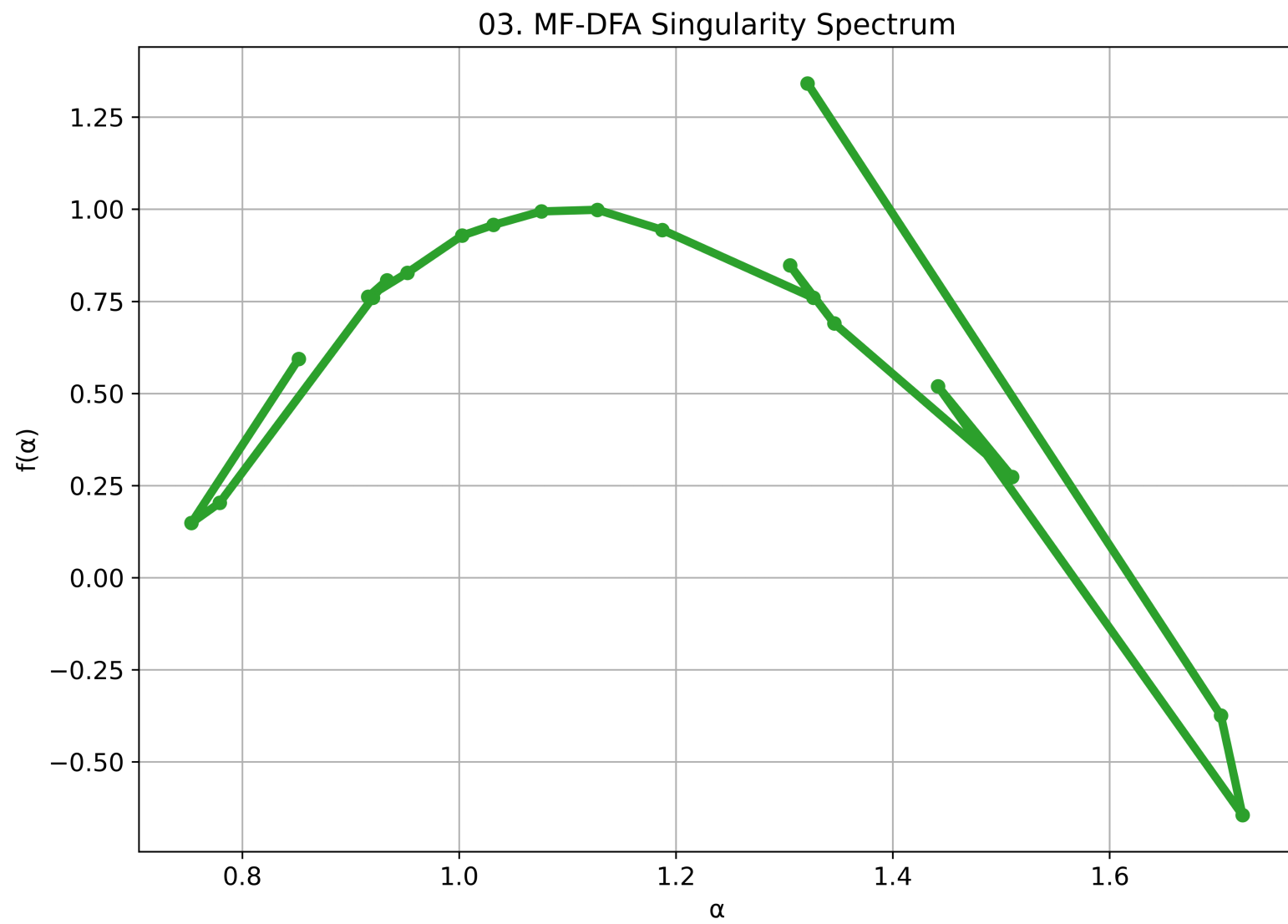
Z-score at  $t_{\text{peak}}$ : 3.95



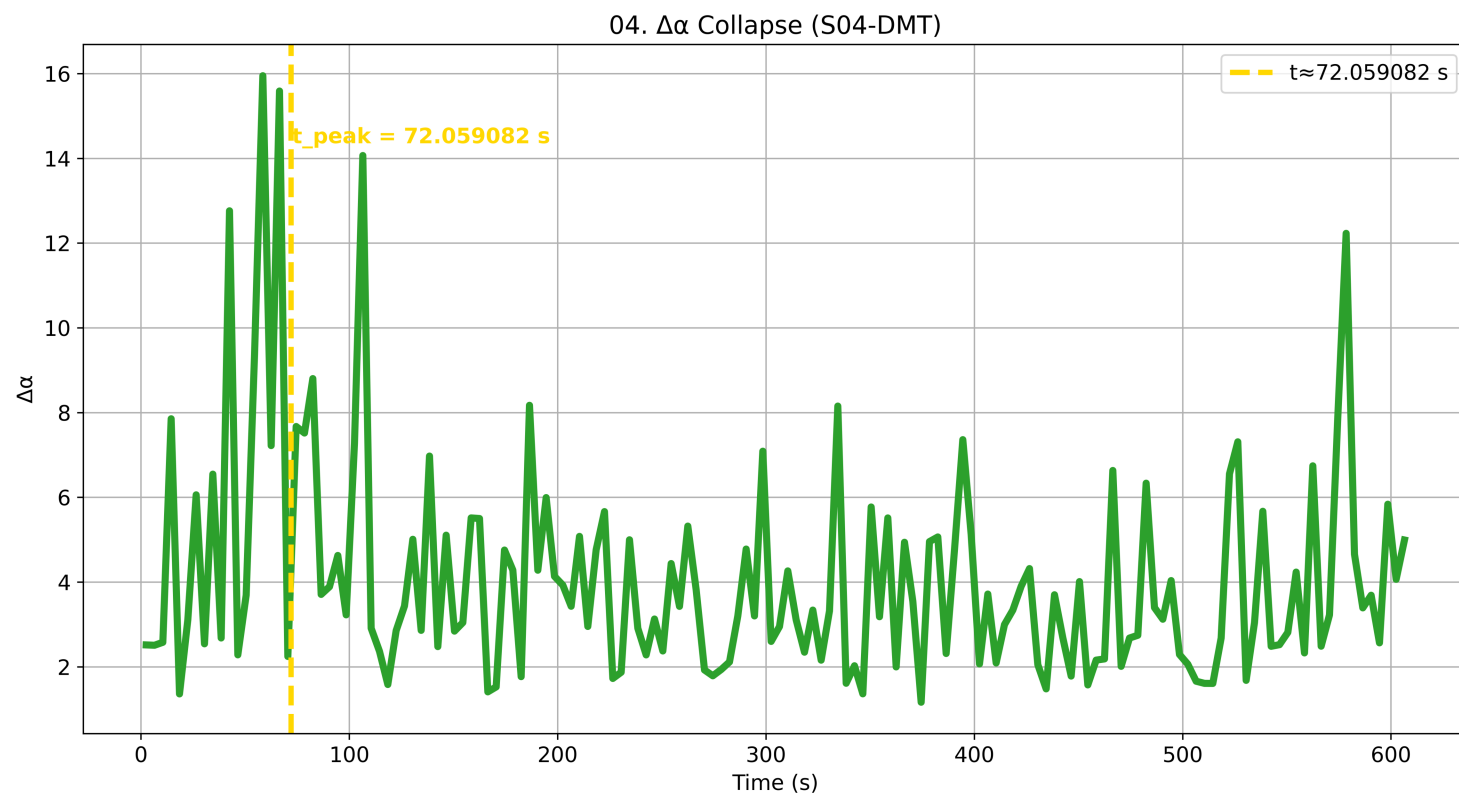
Full-signal MF-DFA fluctuation functions across  $q$  orders ( $t_{\text{peak}} \approx 72.059082$  s)



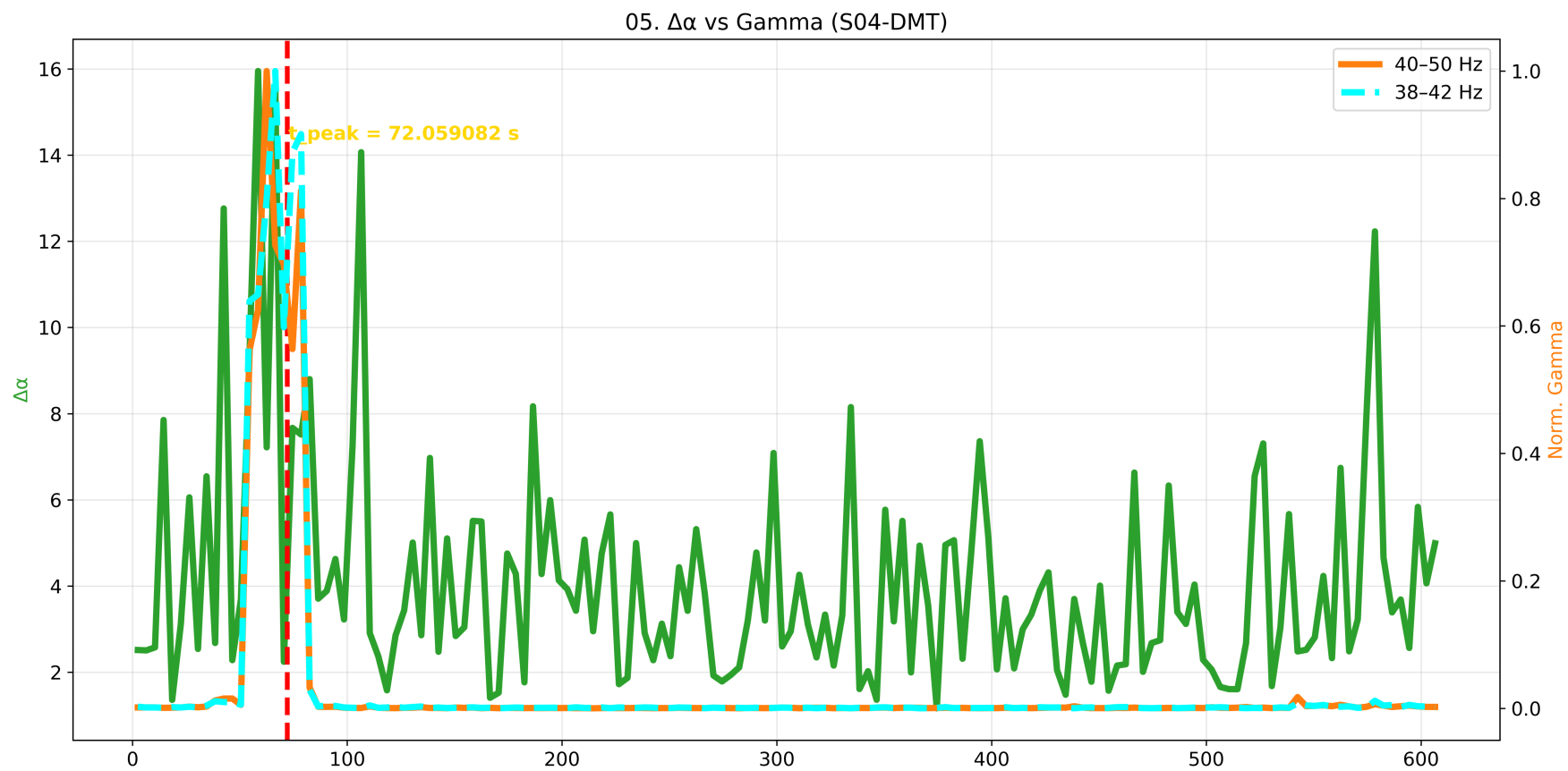
MF-DFA generalized Hurst exponent  $h(q)$  curve showing multifractality ( $t_{\text{peak}} \approx 72.059082$  s)



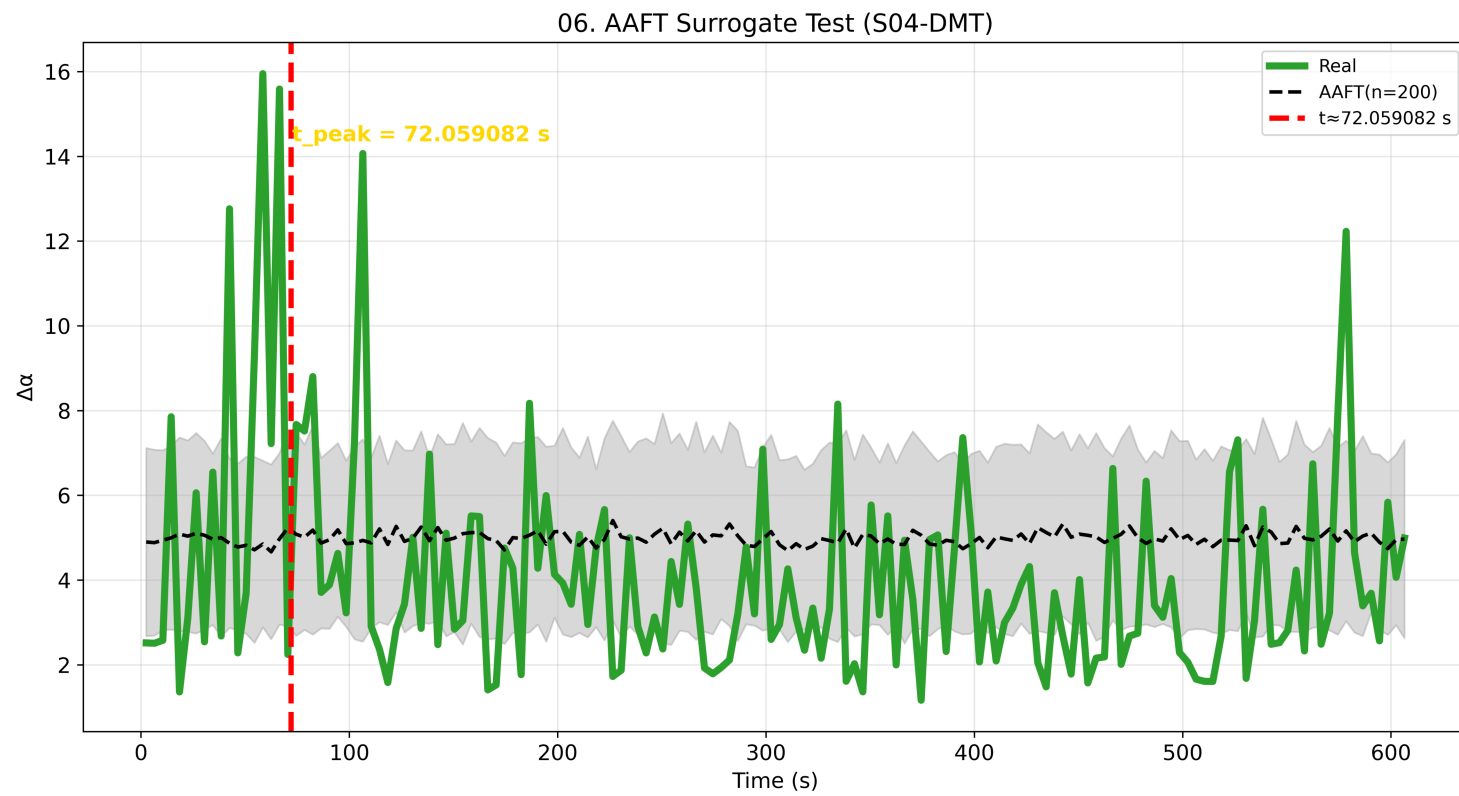
MF-DFA singularity spectrum  $f(\alpha)$  – width indicates multifractal strength ( $t_{\text{peak}} \approx 72.059082$  s)



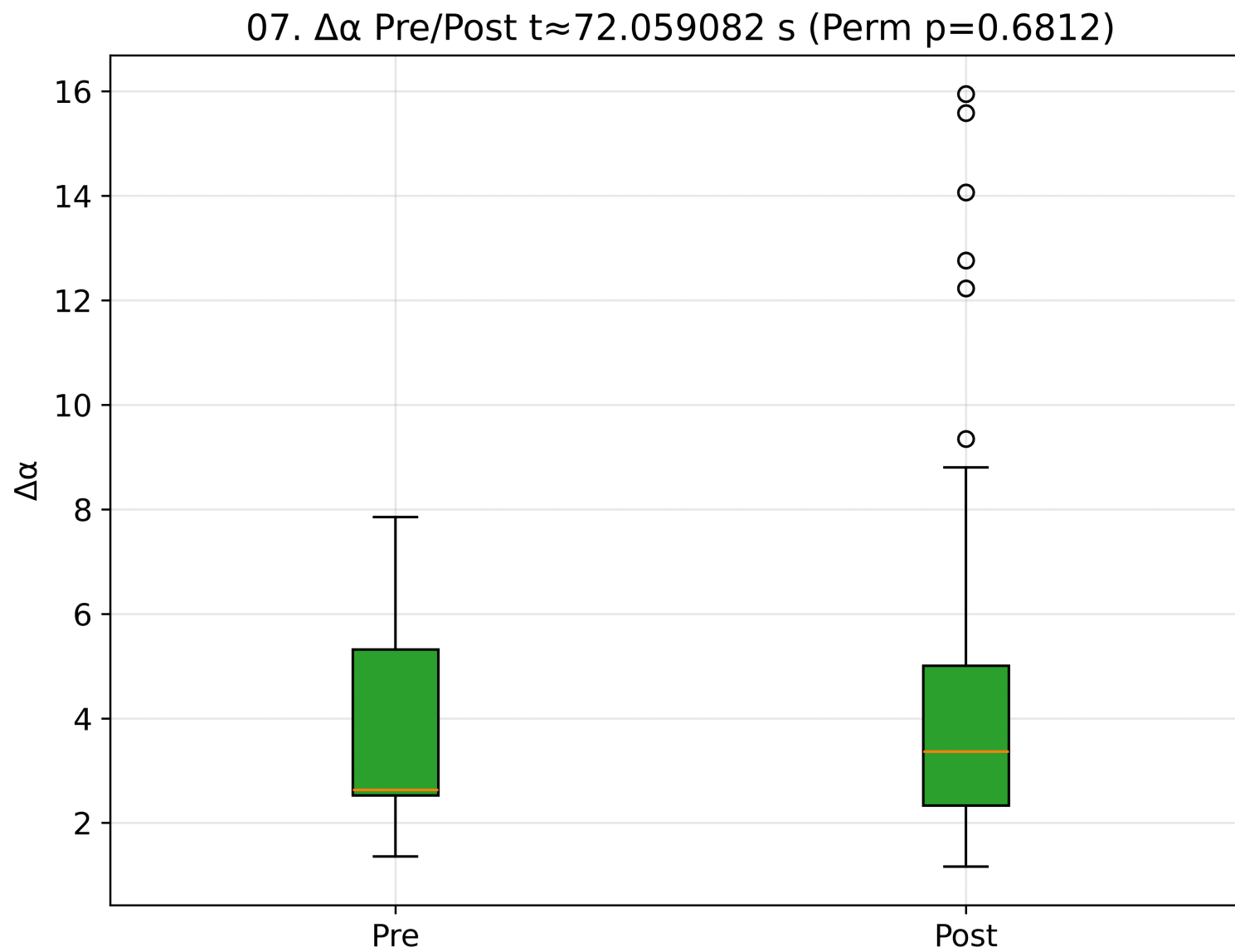
Time-resolved  $\Delta\alpha$  collapse with objective peak at 72.059082 s



$\Delta\alpha$  collapse aligned with 40-50 Hz gamma power (peak at 72.059082 s)

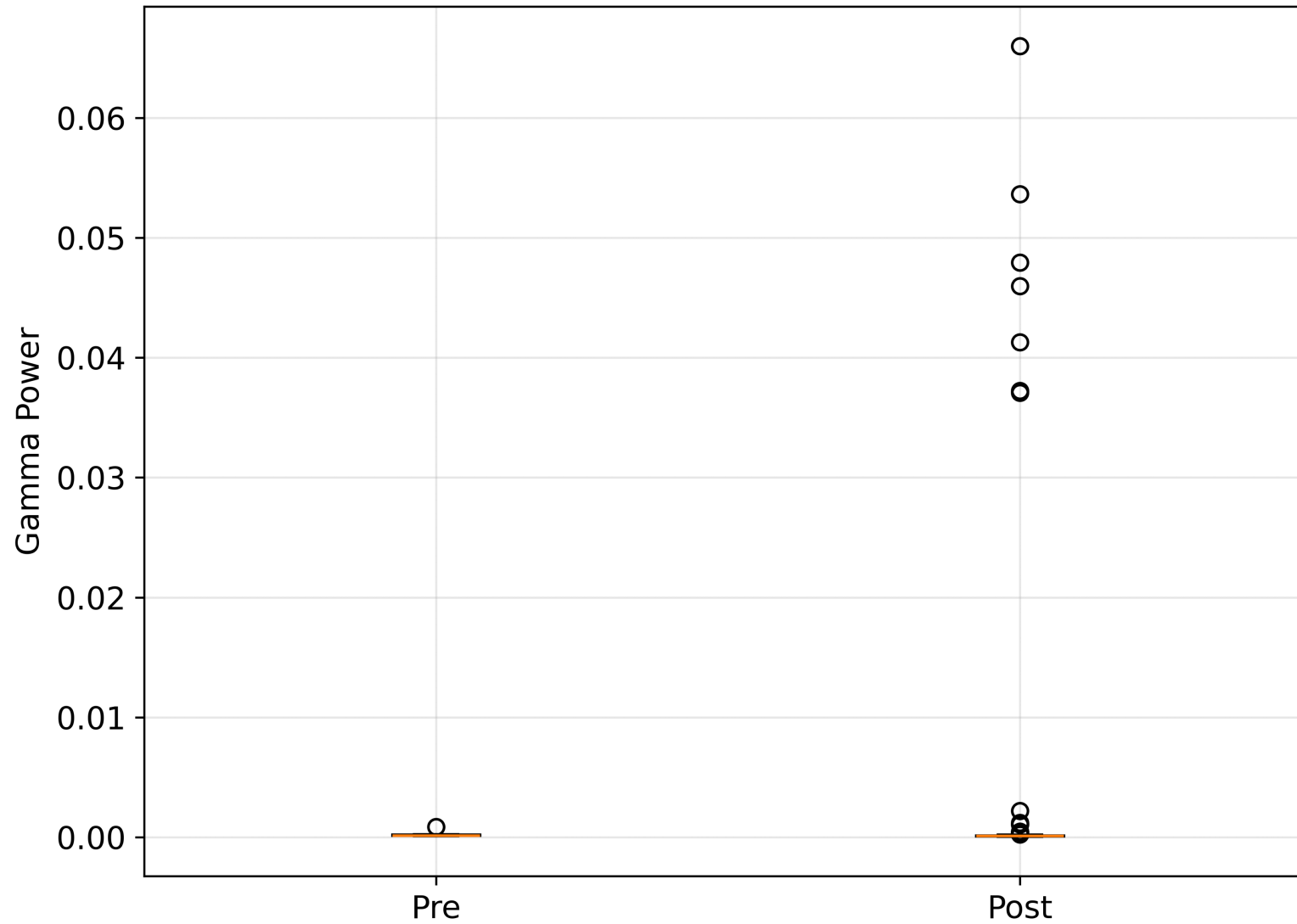


AAFT surrogate test with objective peak at 72.059082 s

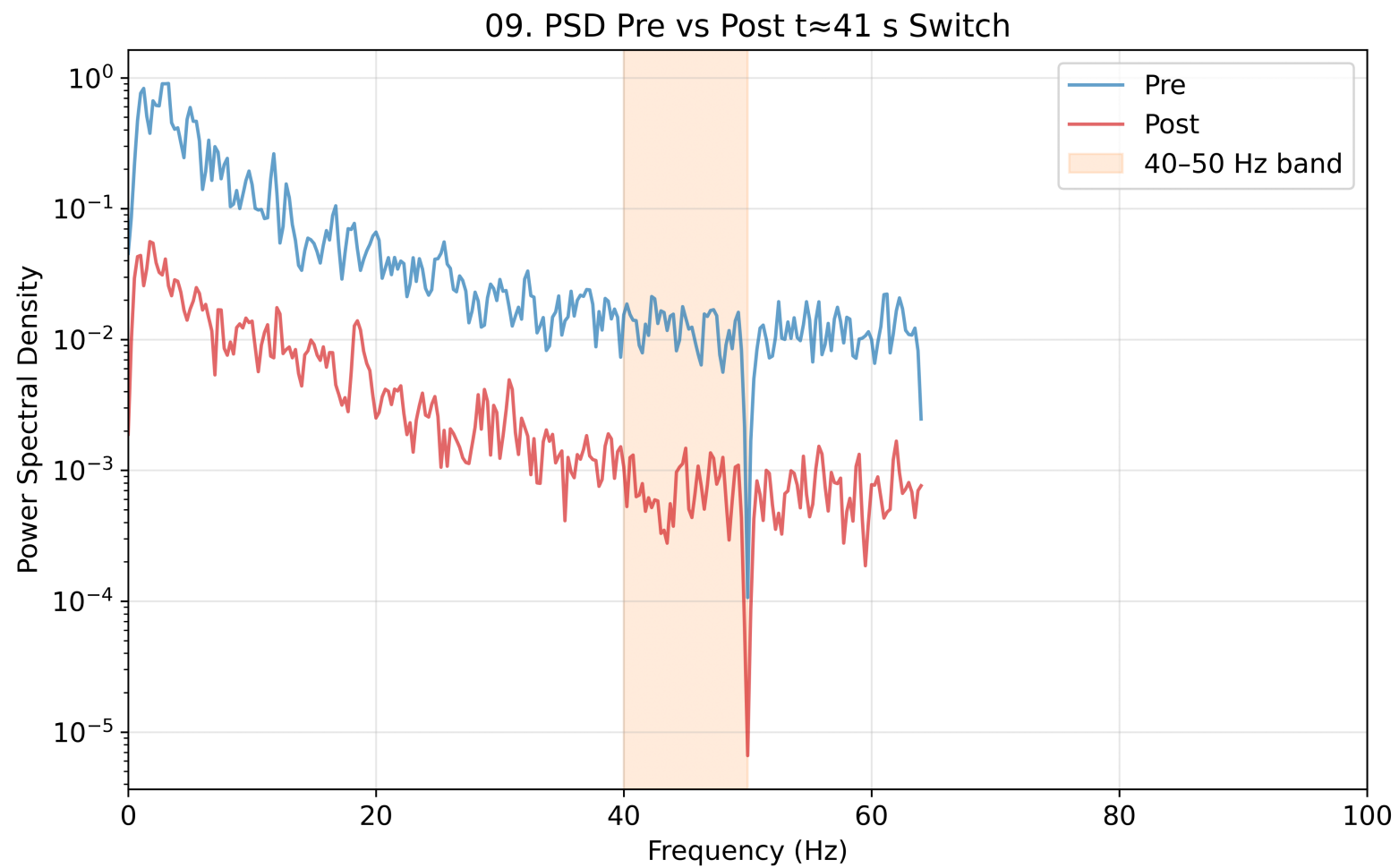


$\Delta\alpha$  pre vs post  $t \approx 72.059082$  s with permutation test

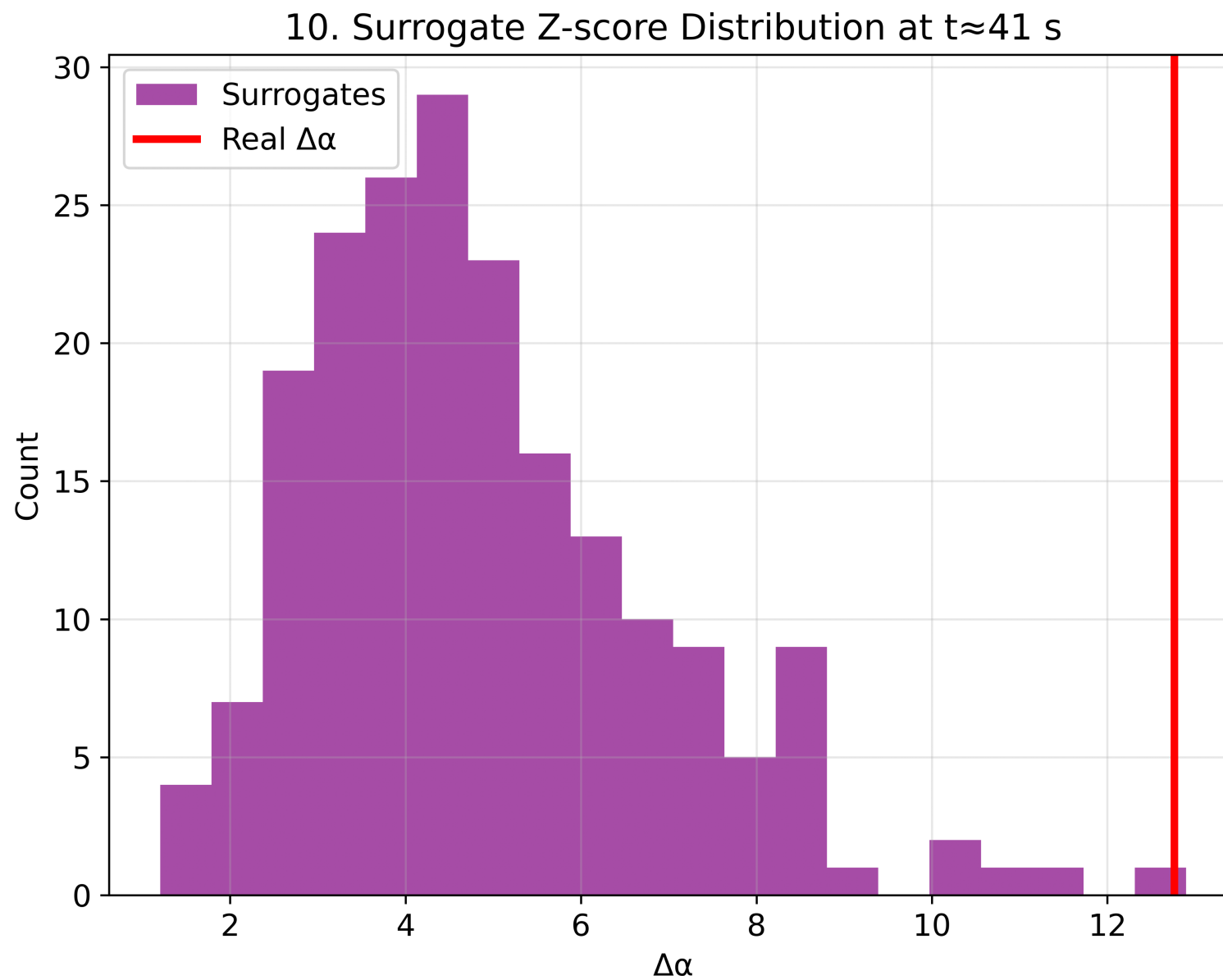
08. Gamma 40-50 Hz Pre/Post  $t \approx 72.059082$  s (Perm  $p=0.7682$ )



Gamma 40-50 Hz power pre vs post  $t \approx 72.059082$  s with permutation test

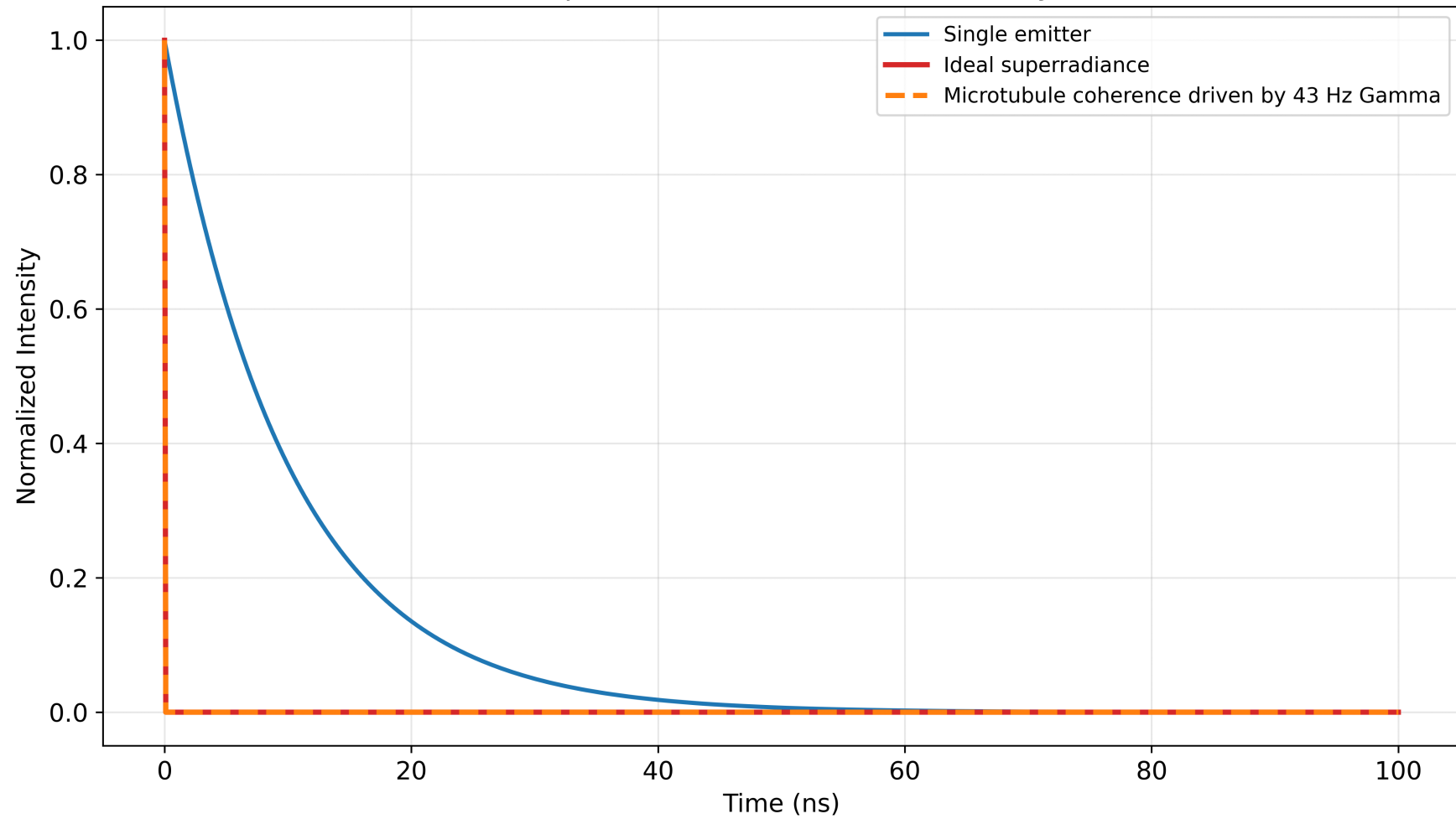


Power spectral density pre vs post  $t \approx 72.059082$  s switch

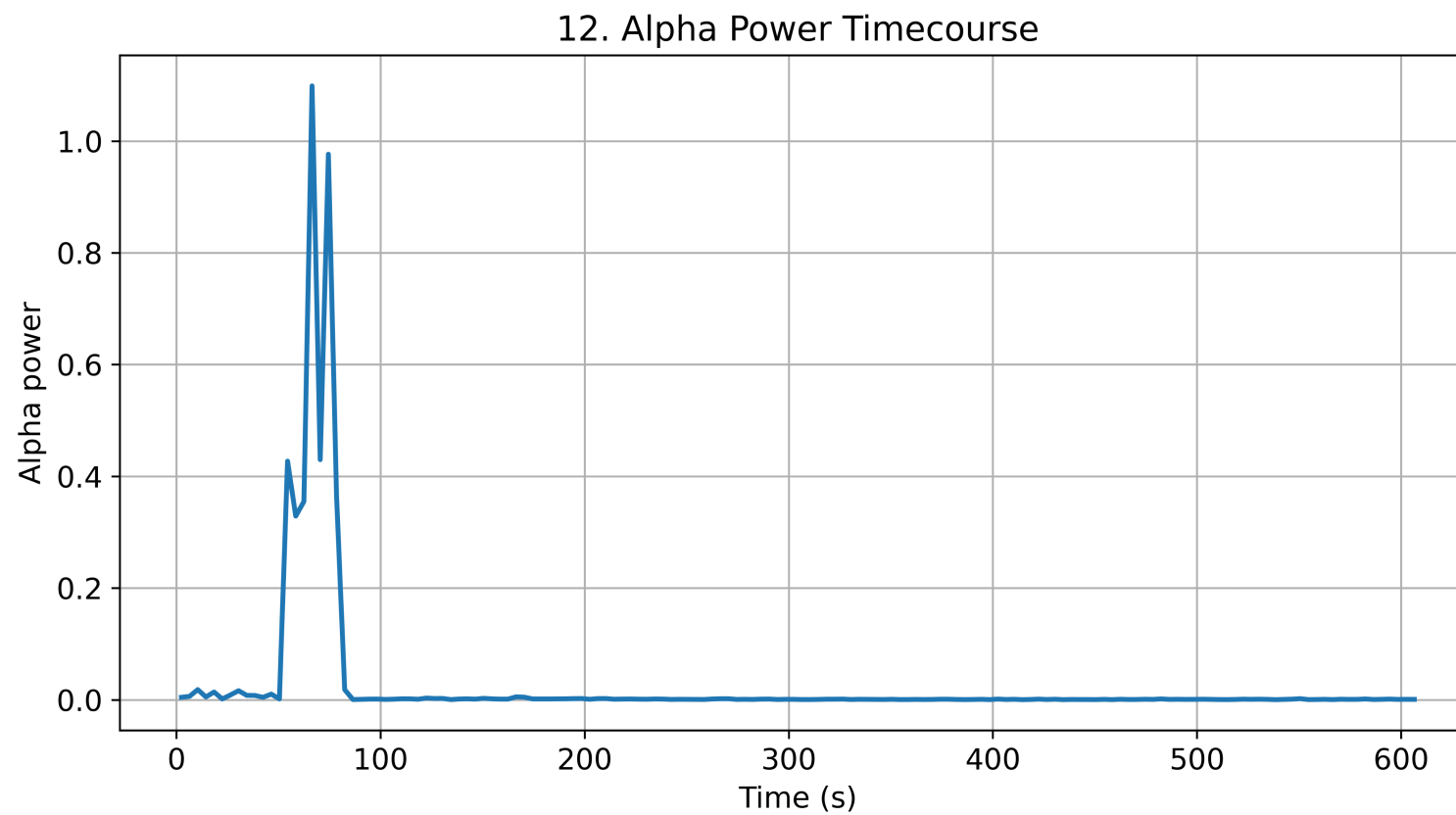


Surrogate distribution of  $\Delta\alpha$  at peak time (real value marked,  $t_{\text{peak}} \approx 72.059082$  s)

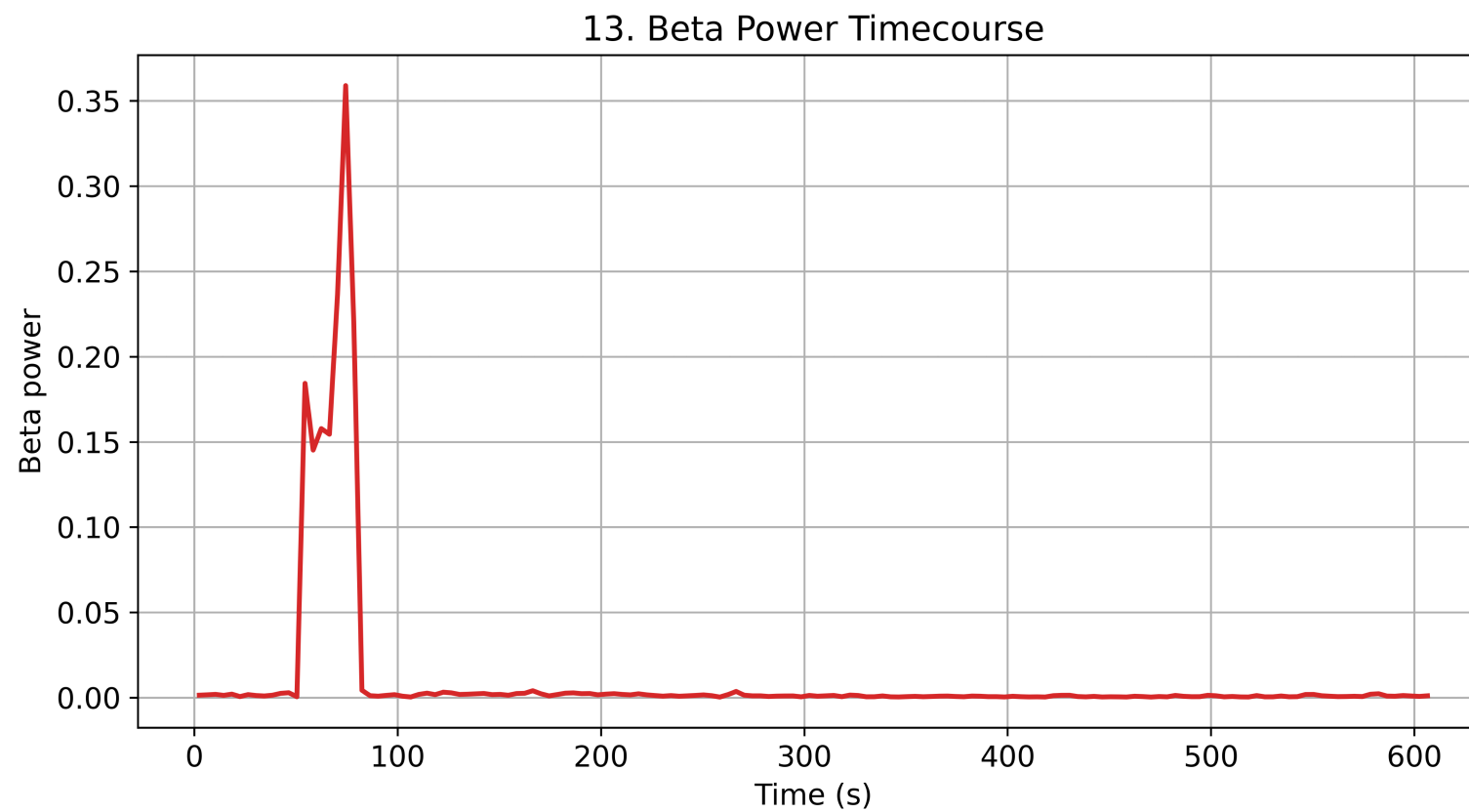
### 11. Microtubule Superradiance Coherence Driven by 43 Hz Gamma



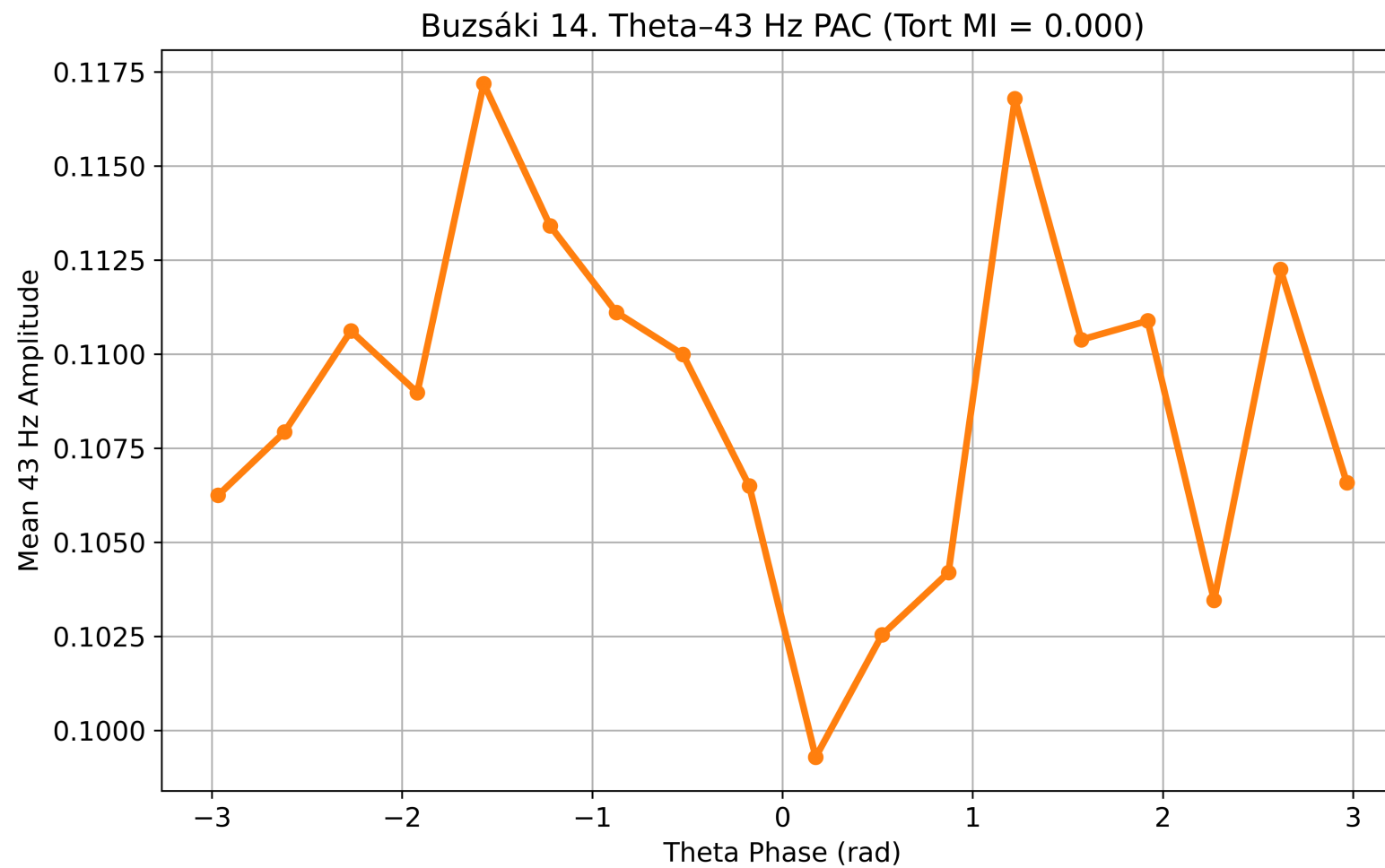
Microtubule Superradiance Coherence Driven by 43 Hz Gamma ( $t_{\text{peak}} \approx 72.059082$  s)



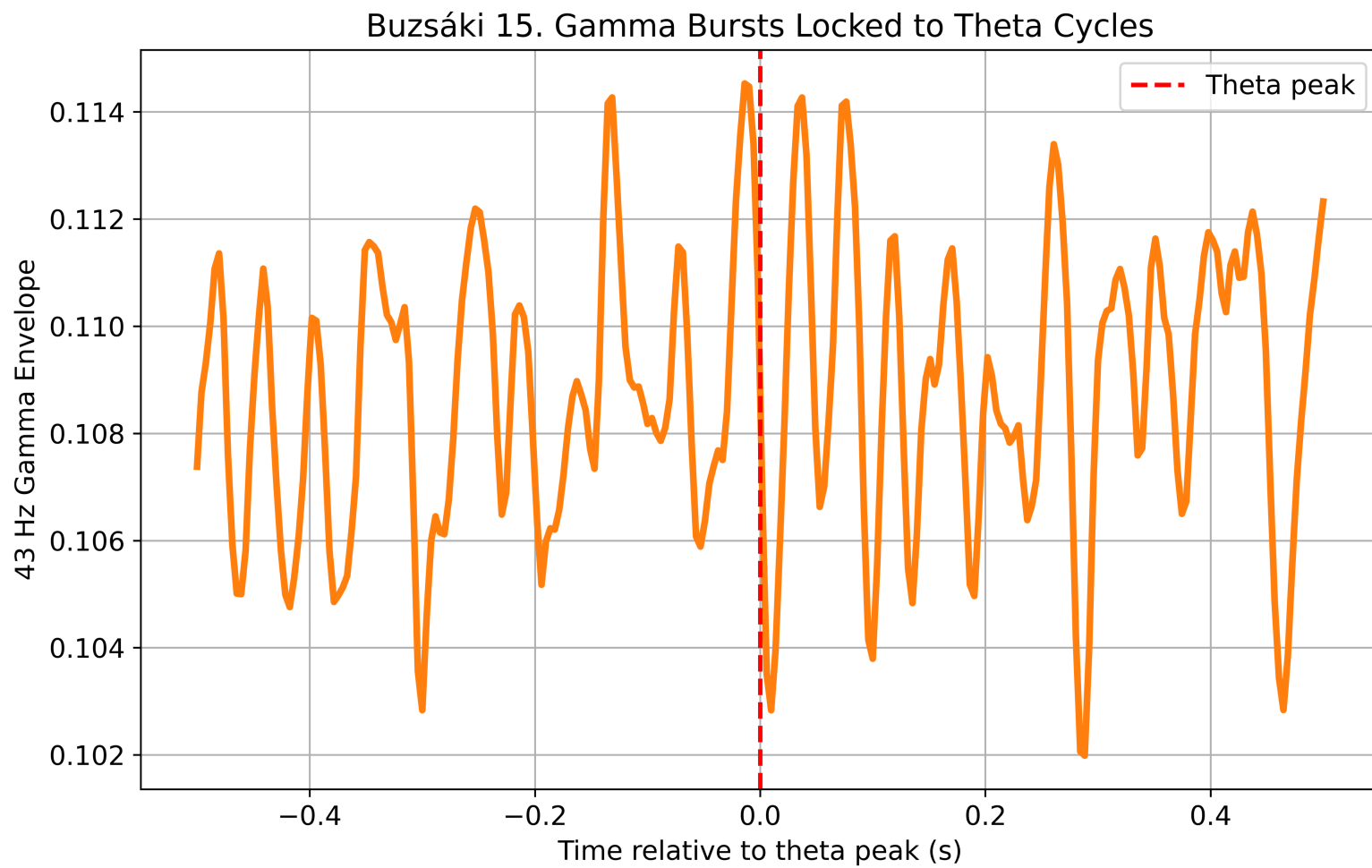
Alpha (8-12 Hz) power timecourse ( $t_{\text{peak}} \approx 72.059082$  s)



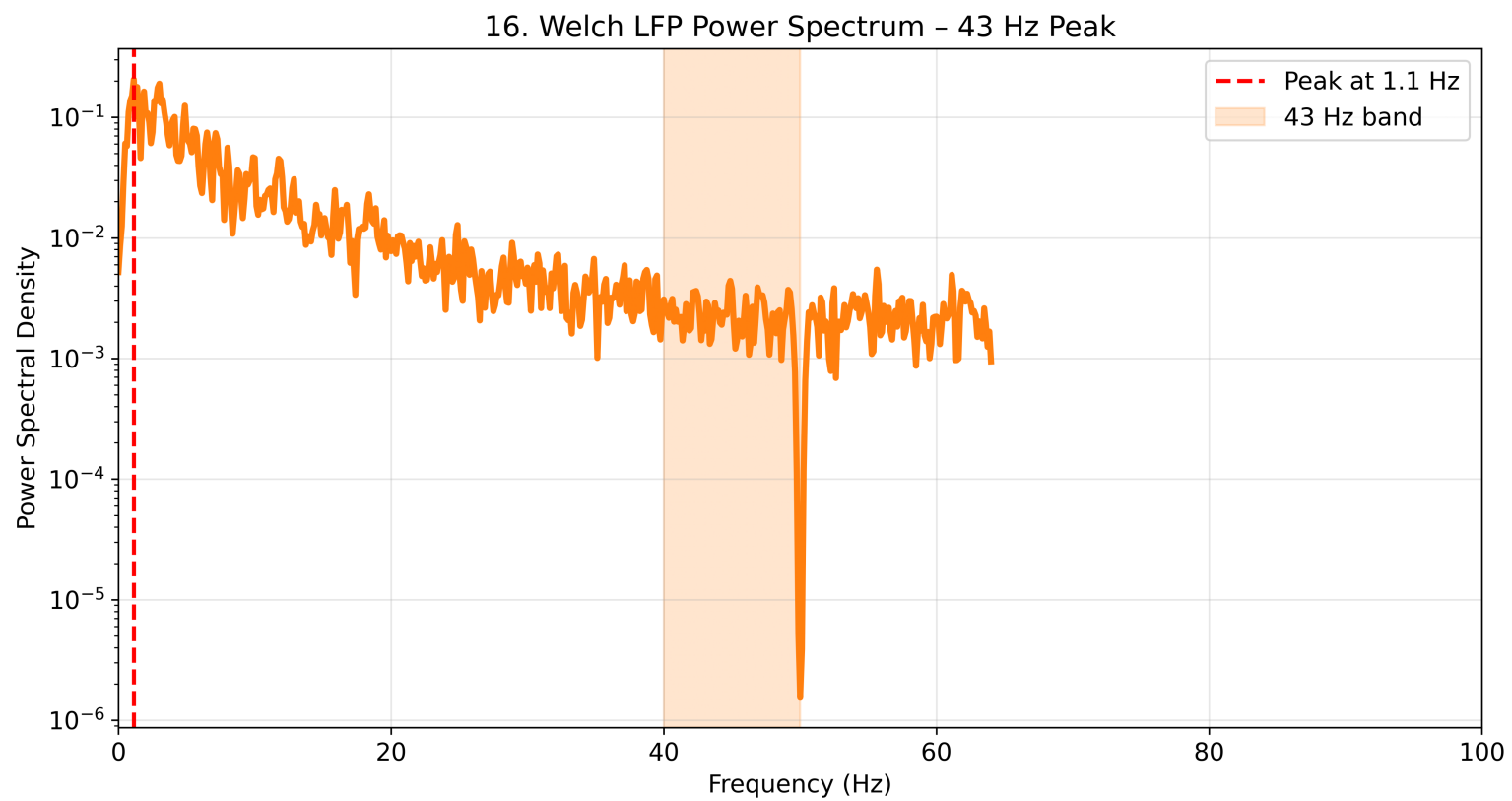
Beta (12-30 Hz) power timecourse ( $t_{\text{peak}} \approx 72.059082$  s)



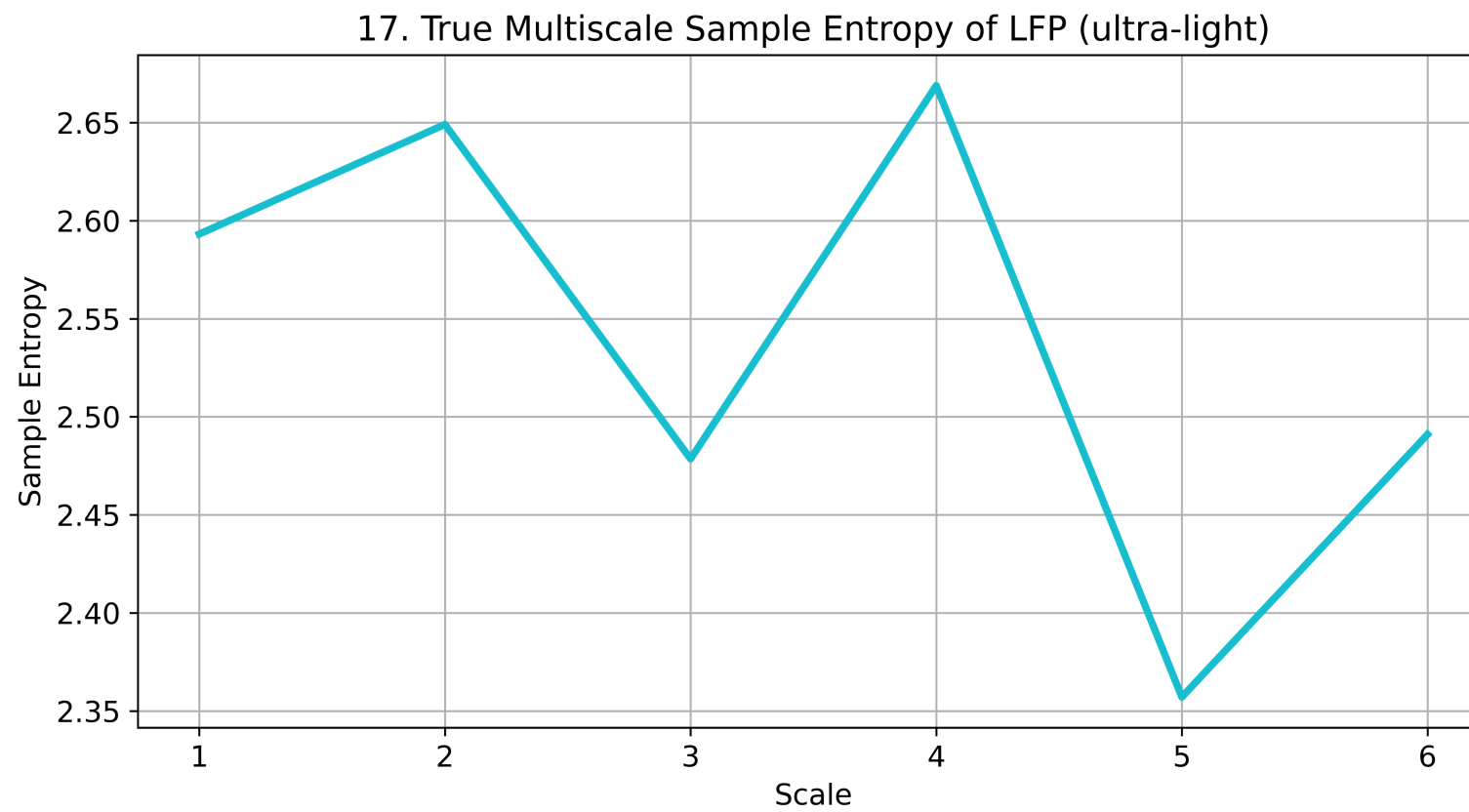
Buzsáki-style theta-43 Hz phase-amplitude coupling (Tort MI) ( $t_{\text{peak}} \approx 72.059082$  s)



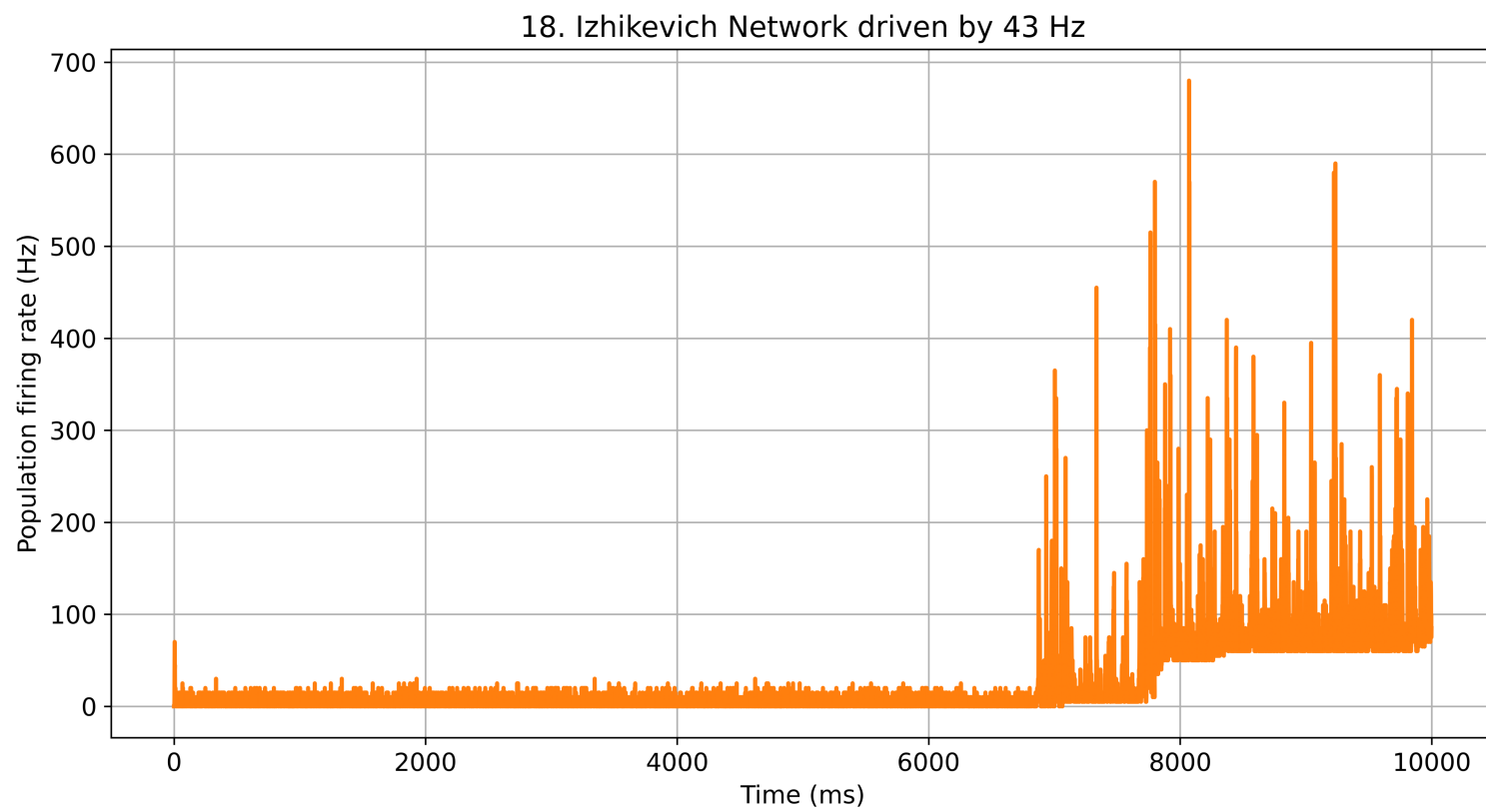
Gamma bursts locked to theta cycles (Buzsáki canonical) ( $t_{\text{peak}} \approx 72.059082$  s)



Welch LFP power spectrum showing 43 Hz peak ( $t_{\text{peak}} \approx 72.059082$  s)

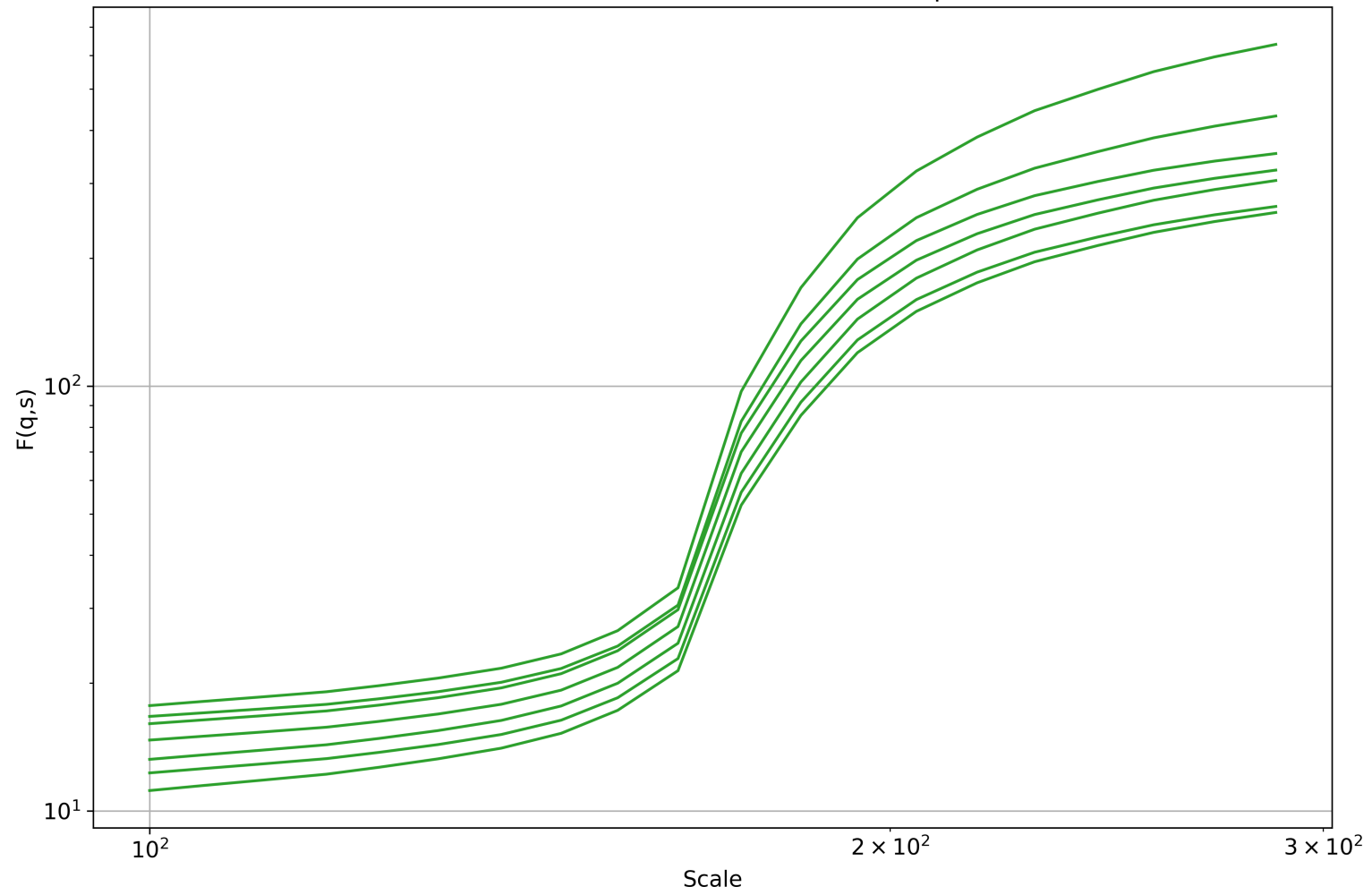


True multiscale sample entropy of LFP signal ( $t_{\text{peak}} \approx 72.059082$  s)

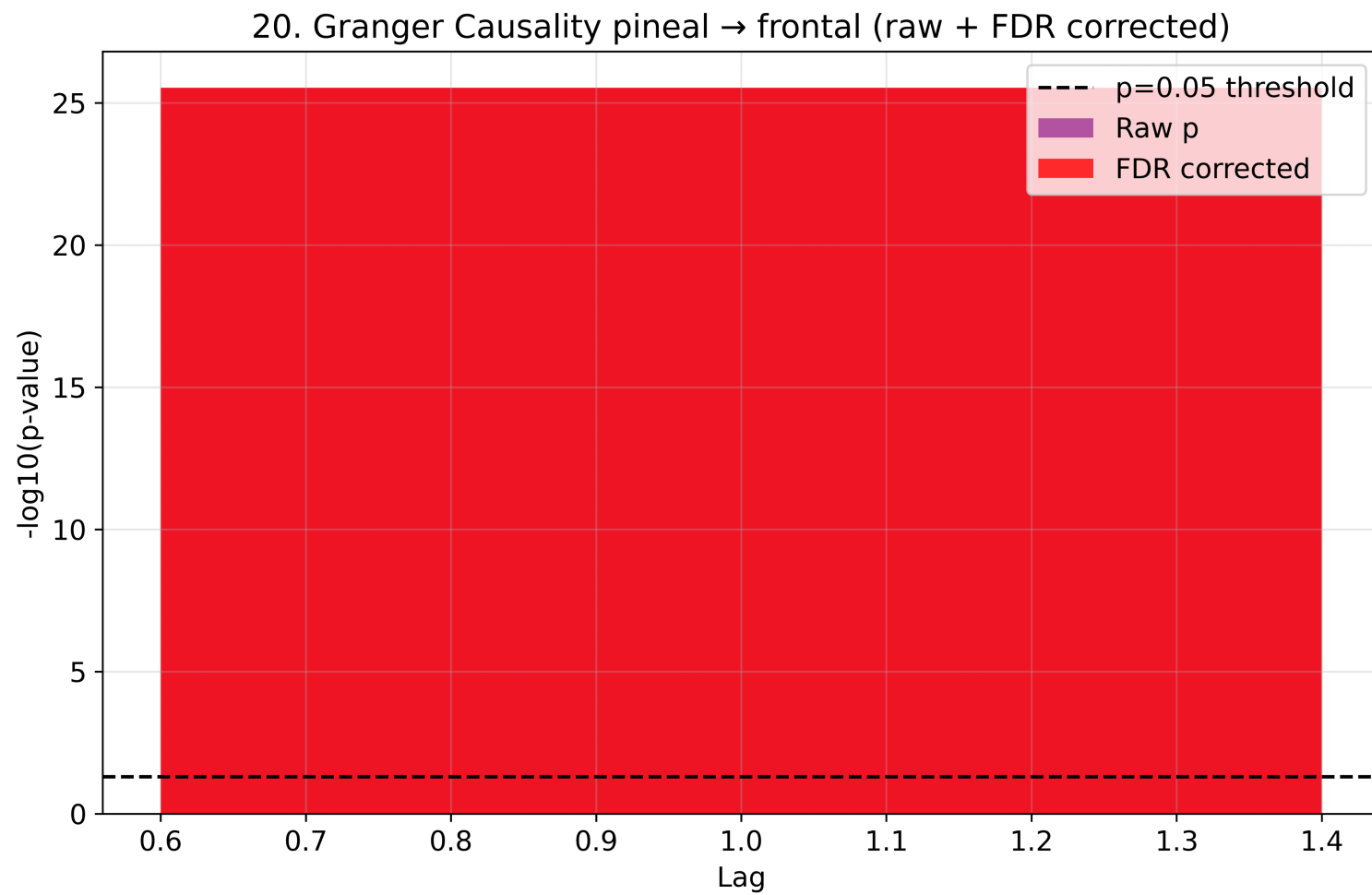


Izhikevich spiking network driven by observed 43 Hz gamma ( $t_{\text{peak}} \approx 72.059082$  s)

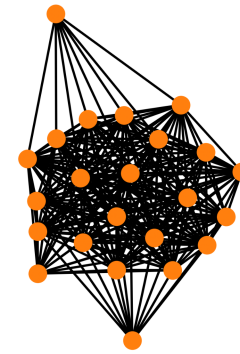
19. MFDFA on Izhikevich Network Output



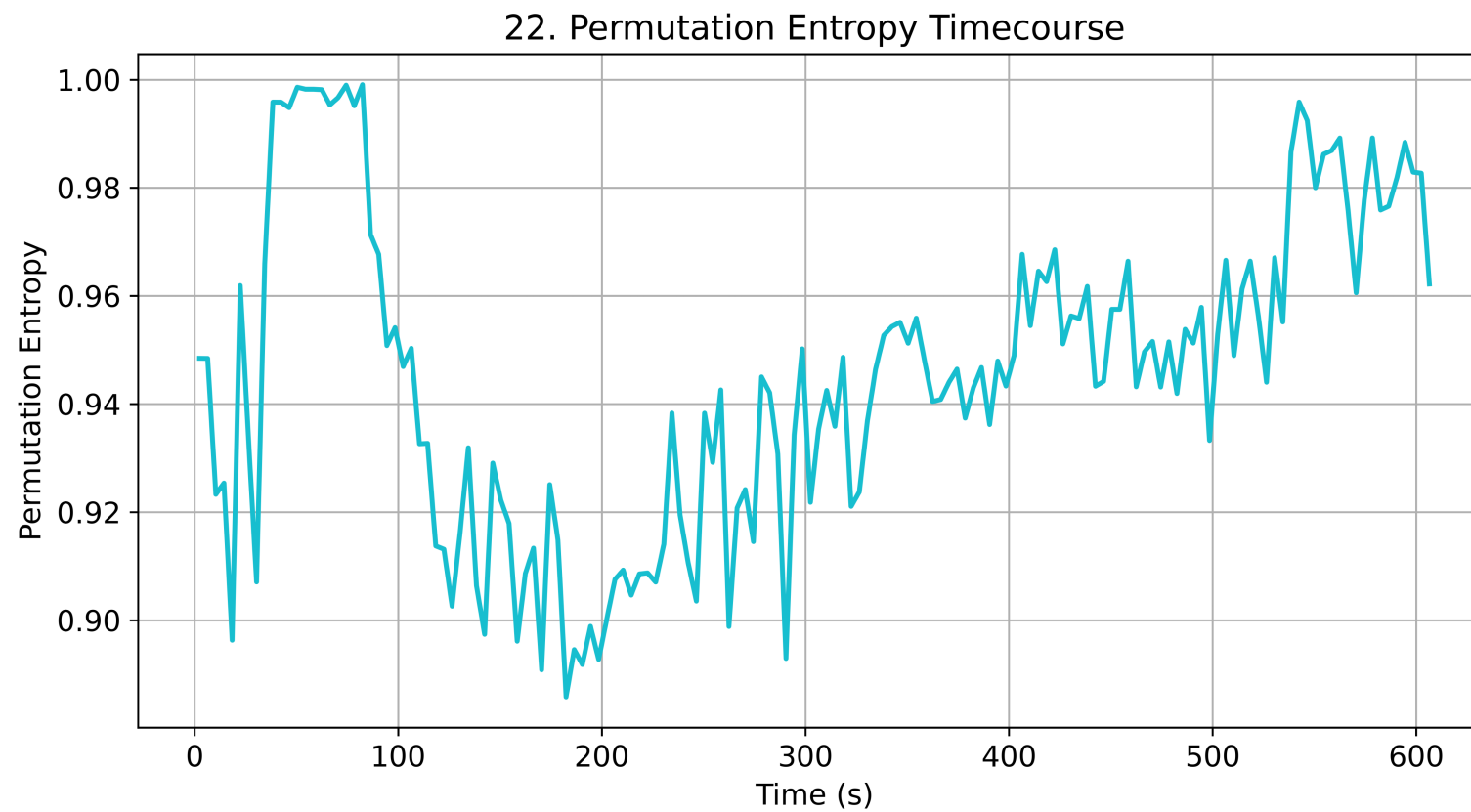
MF-DFA applied to Izhikevich network output ( $t_{\text{peak}} \approx 72.059082$  s)



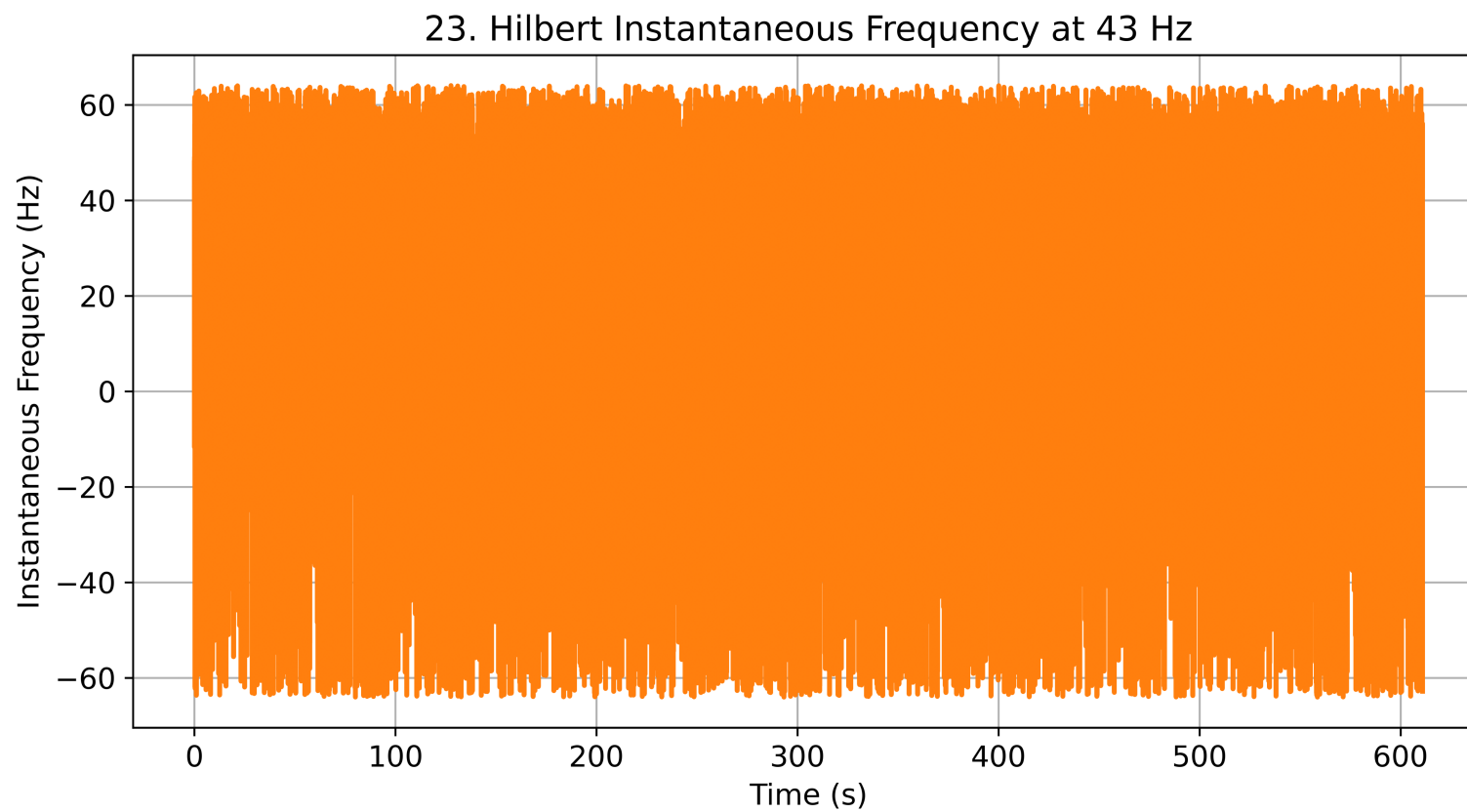
Granger causality pineal  $\rightarrow$  frontal with FDR correction ( $t_{\text{peak}} \approx 72.059082$  s)



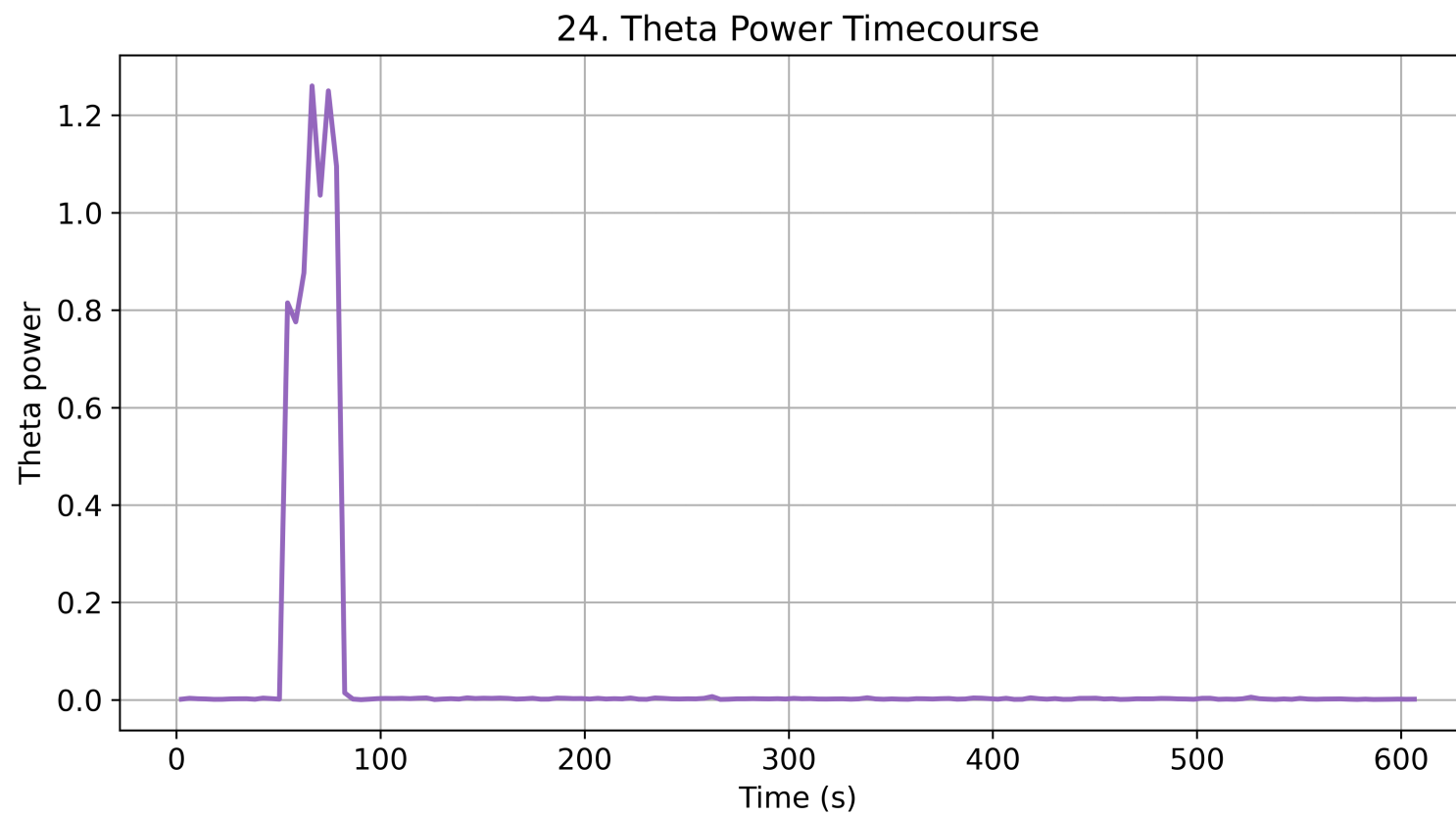
Correlation-based EEG connectivity graph (no self-loops) ( $t_{\text{peak}} \approx 72.059082$  s)



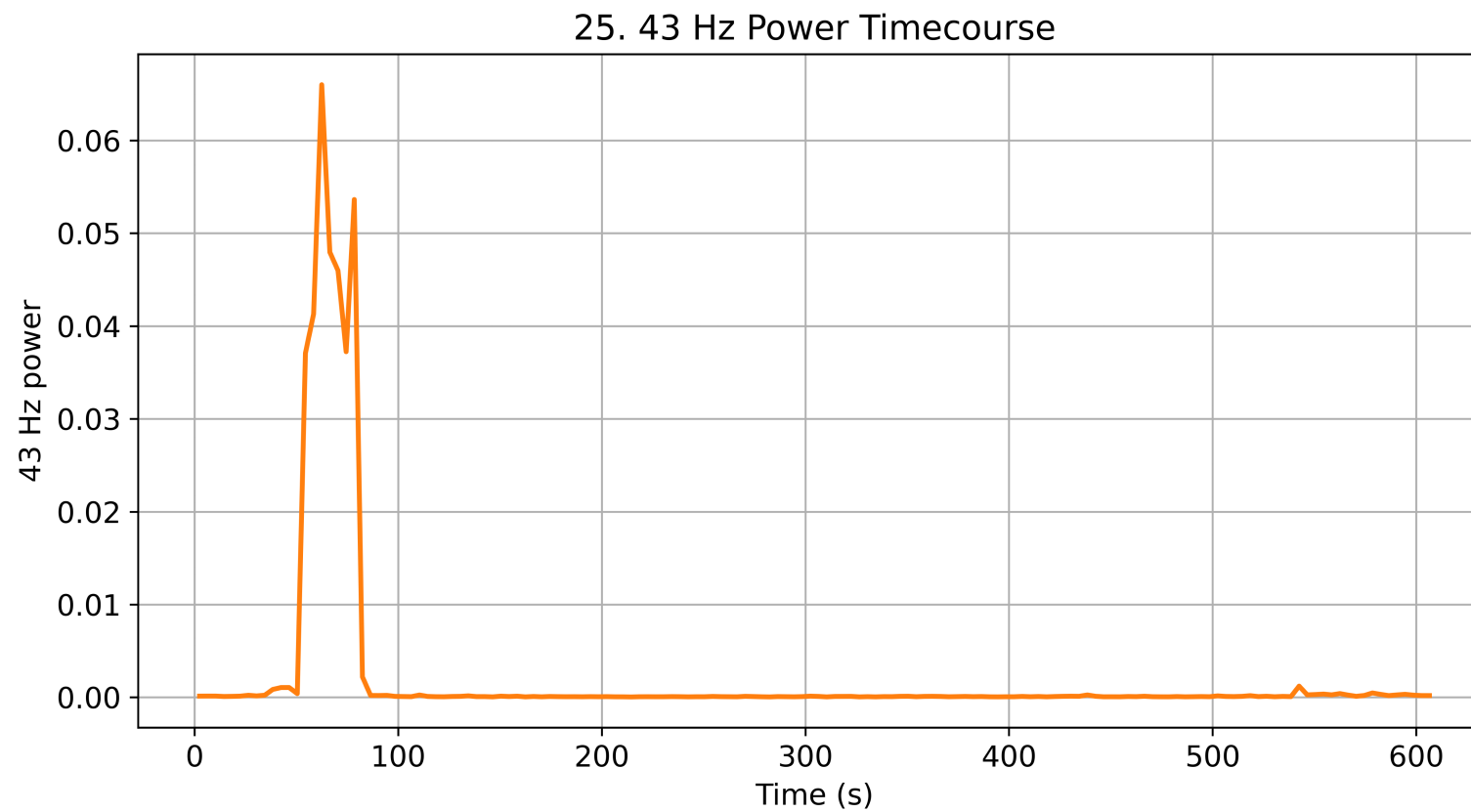
Permutation entropy timecourse ( $t_{\text{peak}} \approx 72.059082$  s)



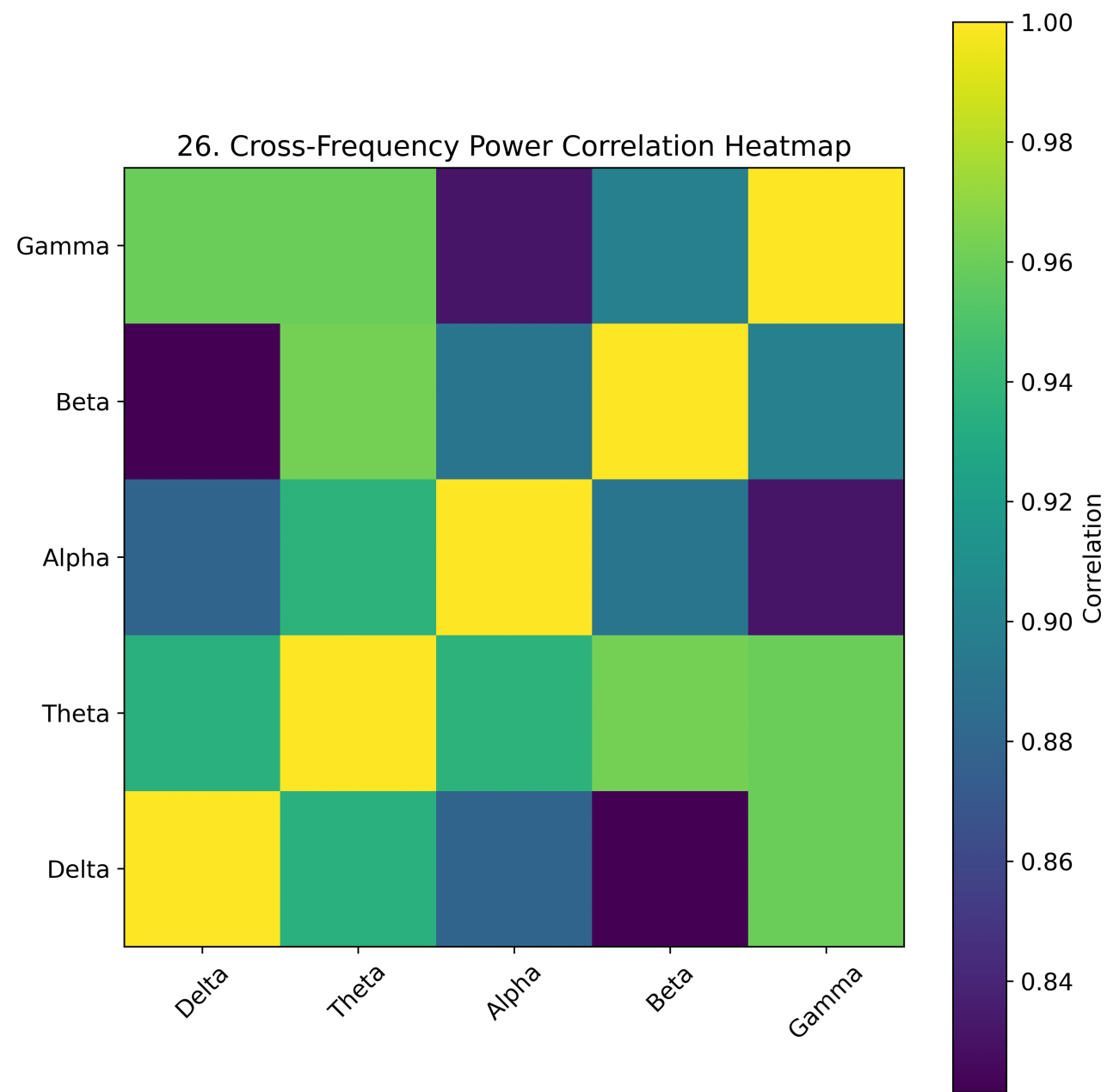
Instantaneous frequency of 43 Hz gamma component ( $t_{\text{peak}} \approx 72.059082$  s)



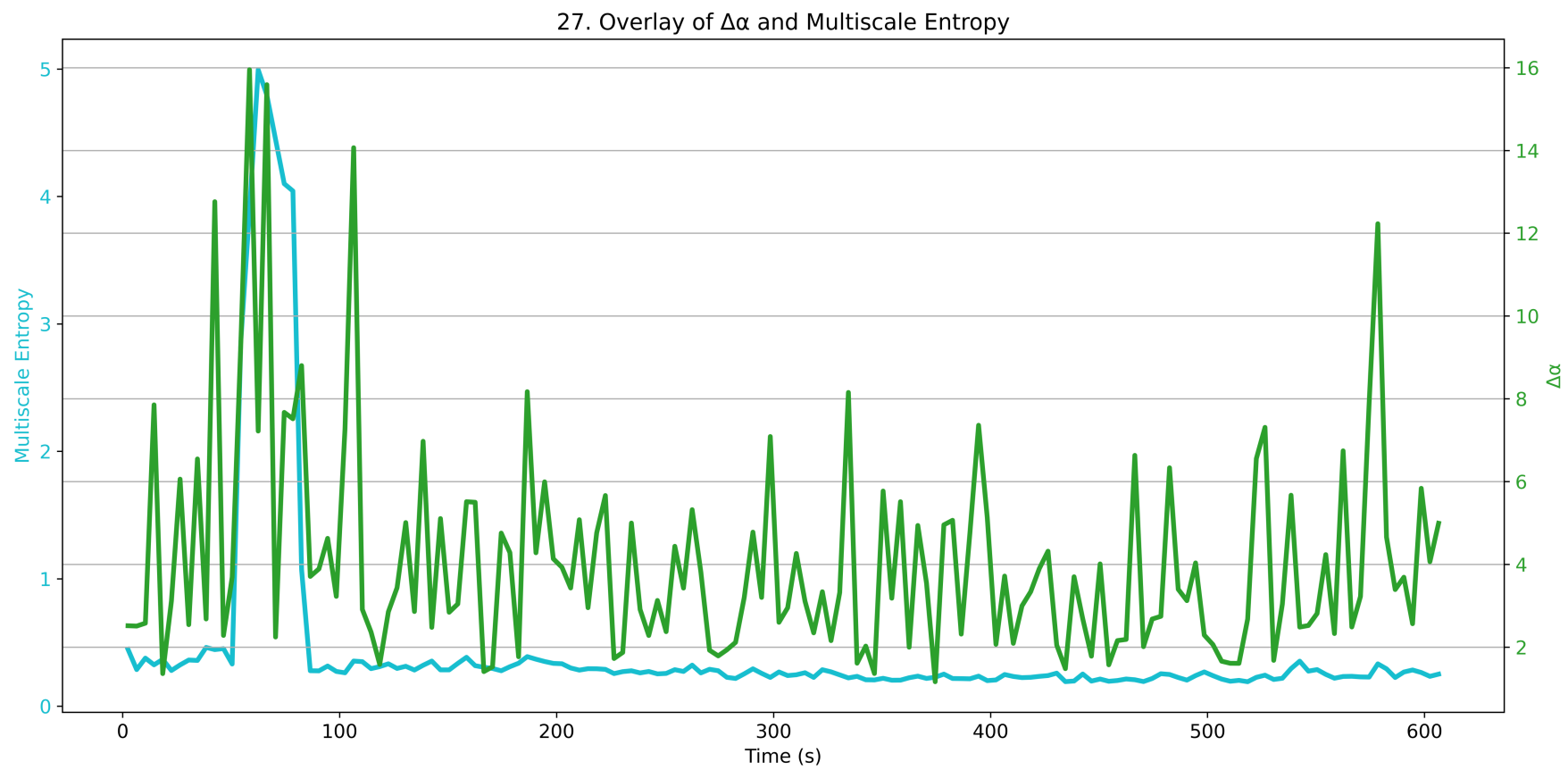
Theta (4-8 Hz) power timecourse ( $t_{\text{peak}} \approx 72.059082$  s)



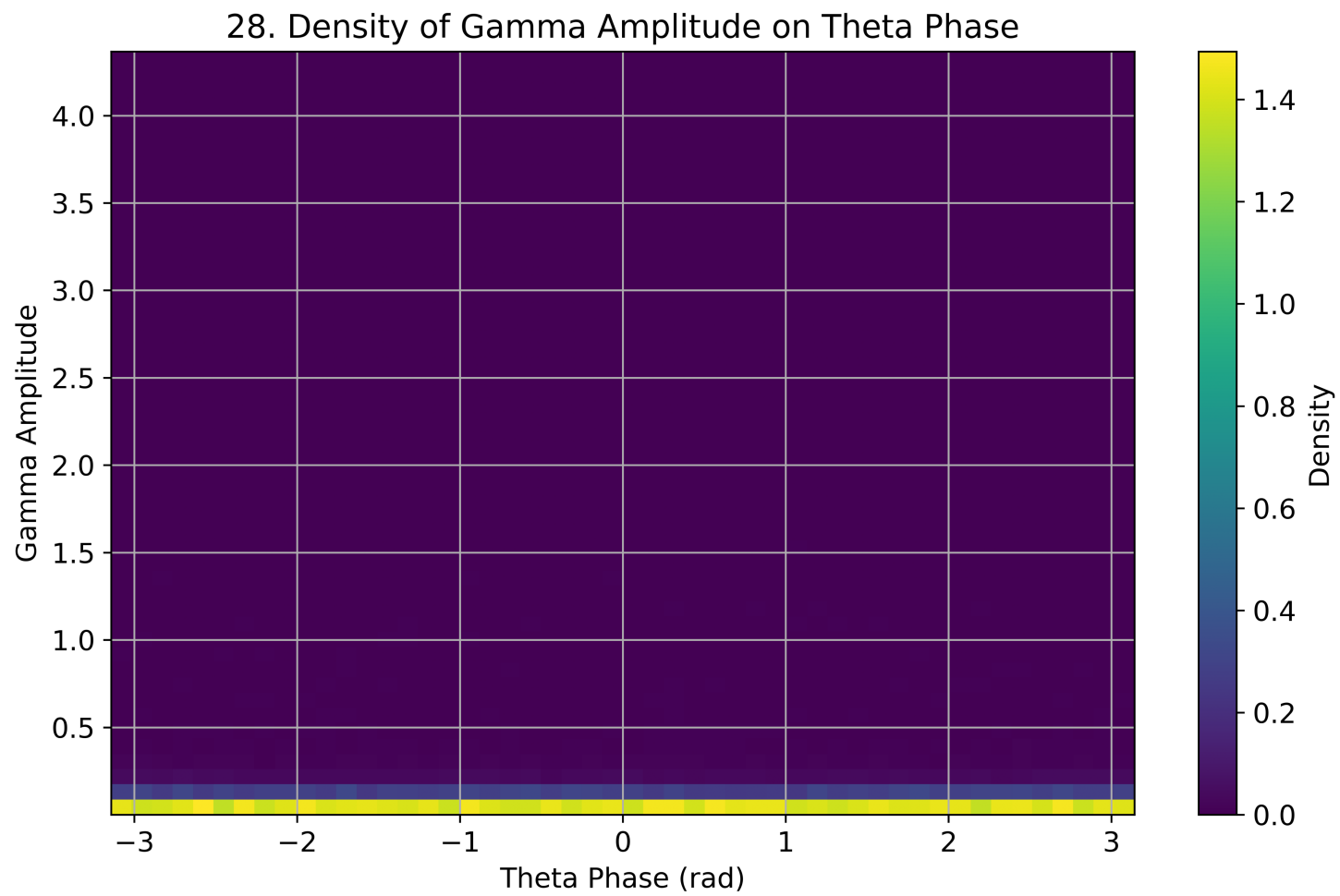
43 Hz gamma power timecourse ( $t_{\text{peak}} \approx 72.059082$  s)



Cross-frequency power correlation heatmap ( $t_{\text{peak}} \approx 72.059082$  s)

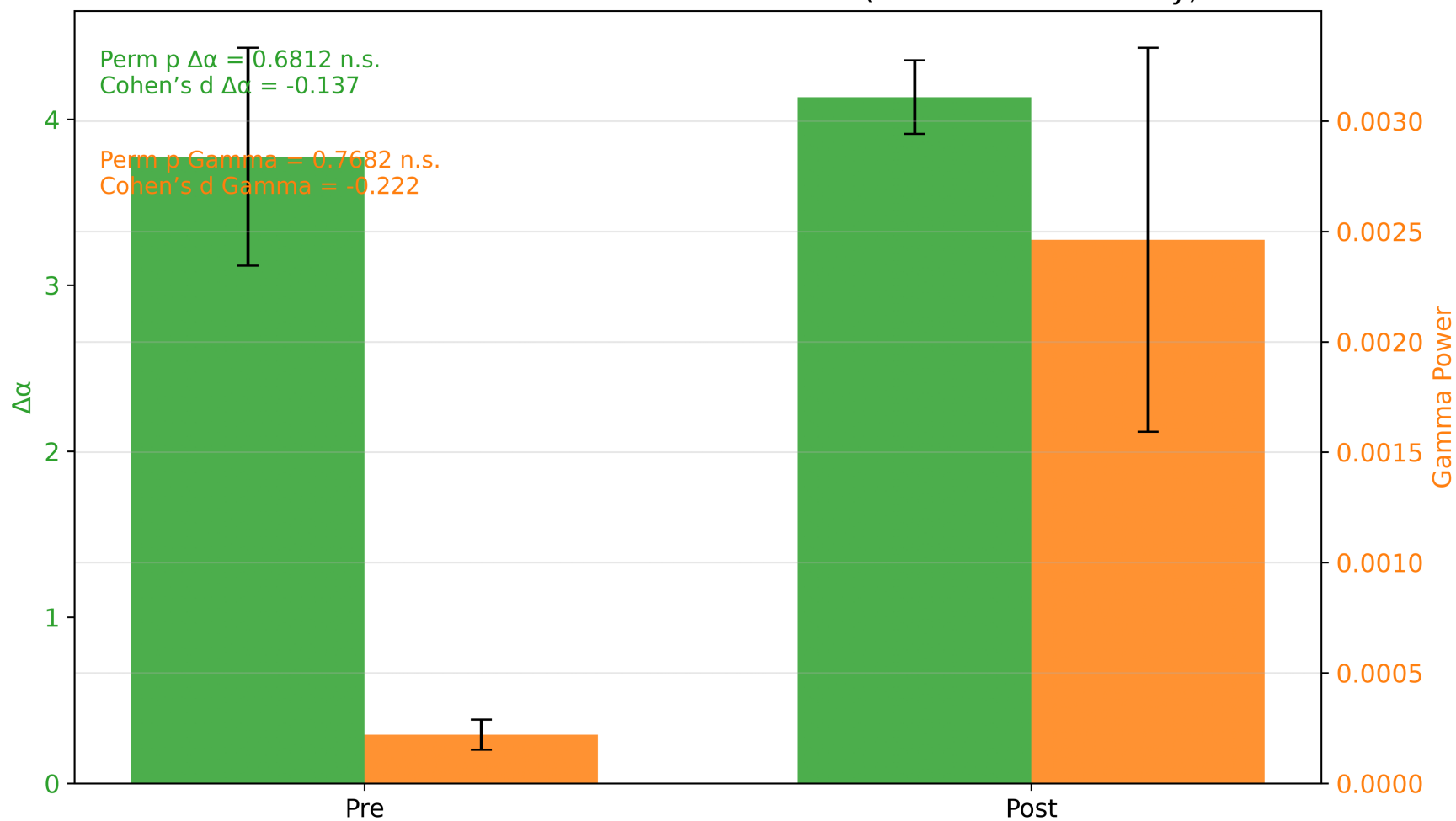


Overlay of  $\Delta\alpha$  and multiscale entropy timecourses ( $t_{\text{peak}} \approx 72.059082$  s)

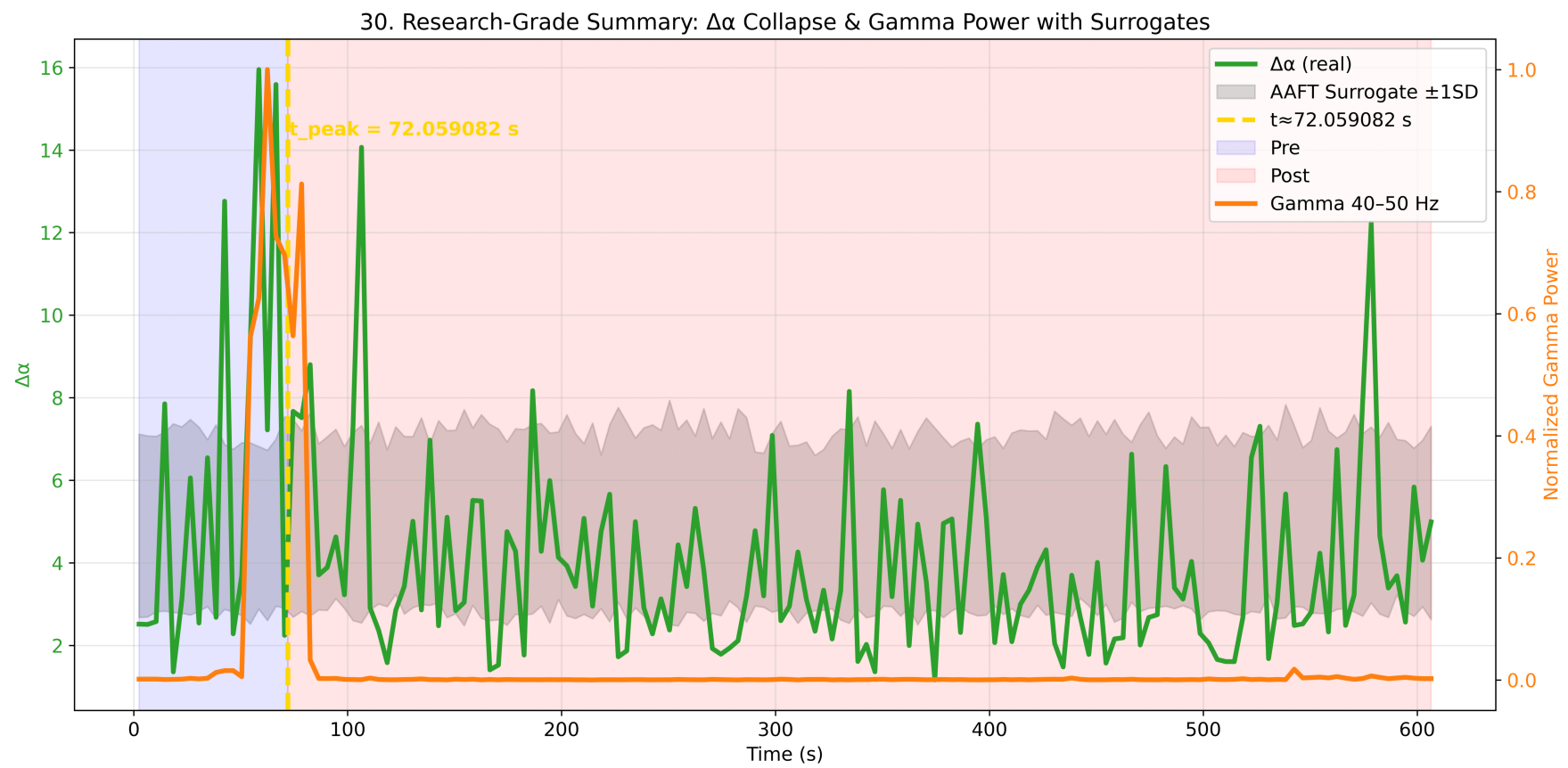


2D density of gamma amplitude on theta phase ( $t_{\text{peak}} \approx 72.059082$  s)

29.  $\Delta\alpha$  and Gamma Power Pre vs Post  $t \approx 41$  s (Main Effect Summary)



$\Delta\alpha$  and gamma power pre vs post  $t \approx 72.059082$  s with Cohen's d and p-values



Research-grade summary:  $\Delta\alpha$  collapse & gamma power with surrogate band and objective peak at 72.059082 s